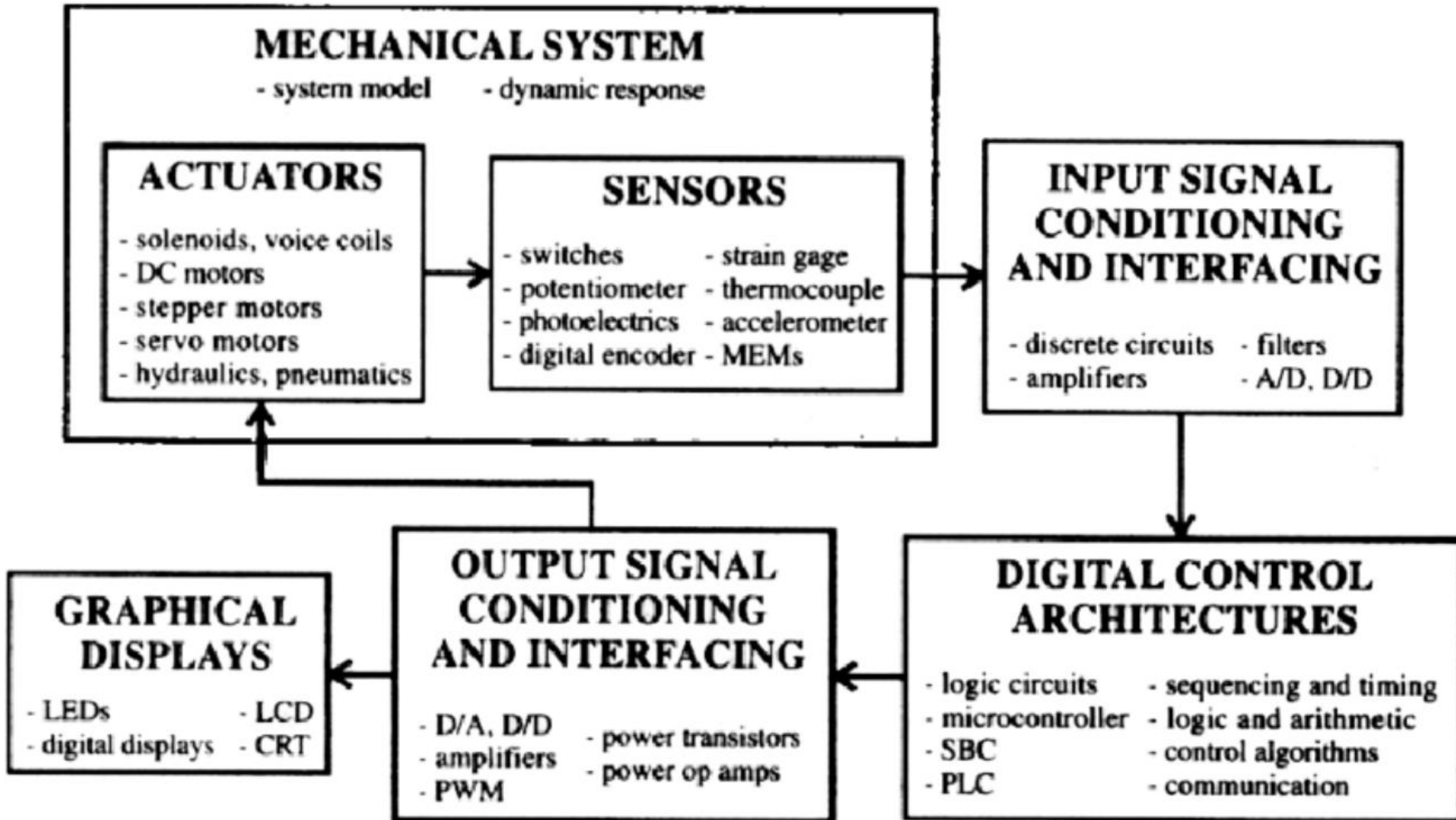


19-205-0403:MECHATRONICS

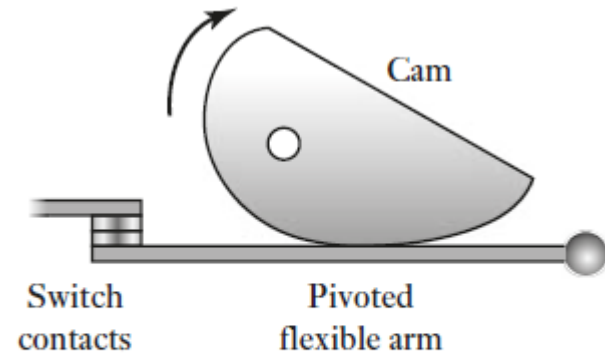
# Designing of Mechatronic Systems

# Key elements of Mechatronics system



# Stages in designing mechatronic systems

- Consider a simple requirement for a device which switches on some actuator e.g. a motor, for some prescribed time
- Possible solutions are
  - Rotating cam
  - Programmable logic controller (PLC)
  - Microprocessor
  - Microcontroller
  - a timer

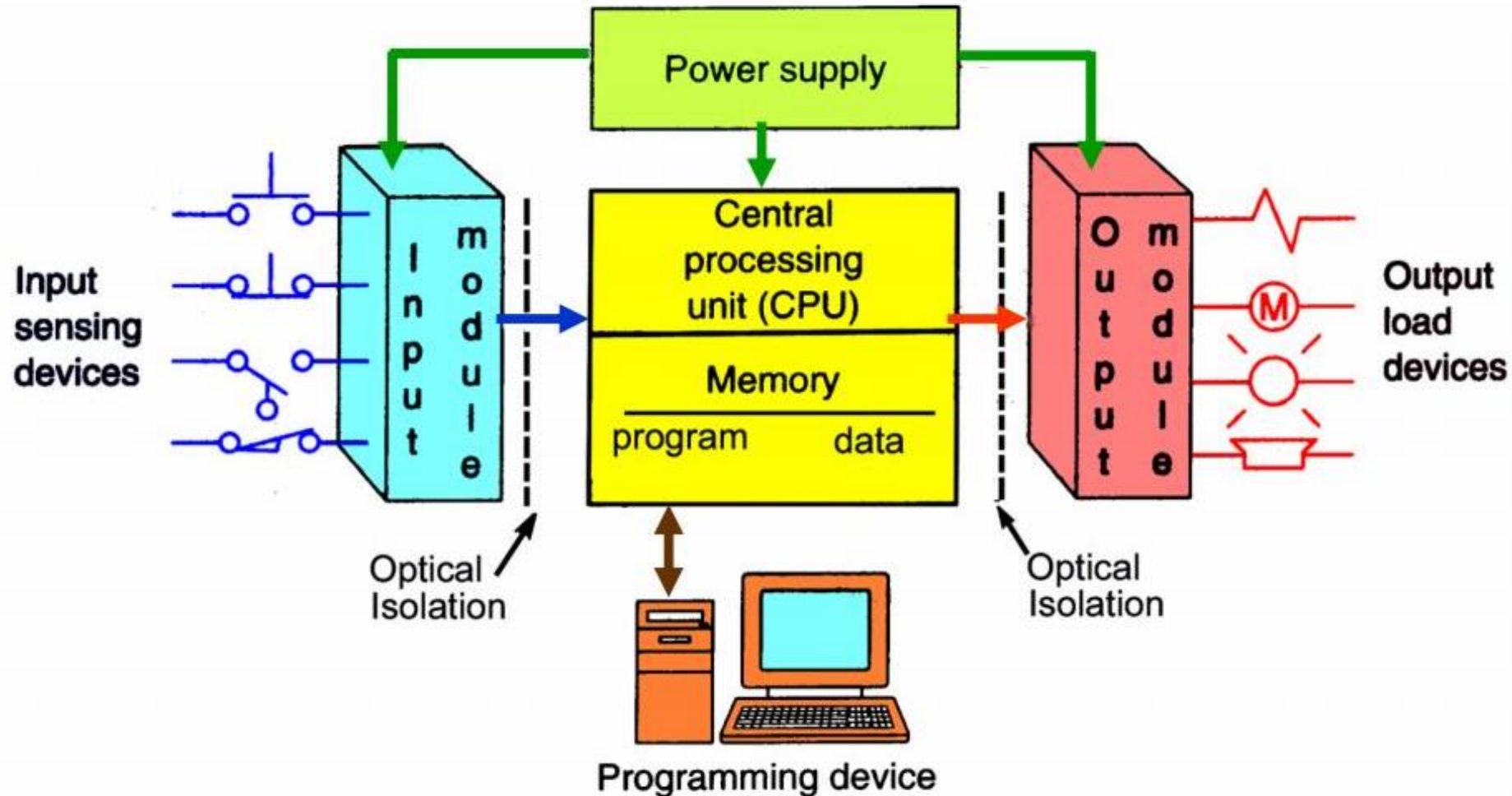


**Figure 24.1** Cam-operated switch.

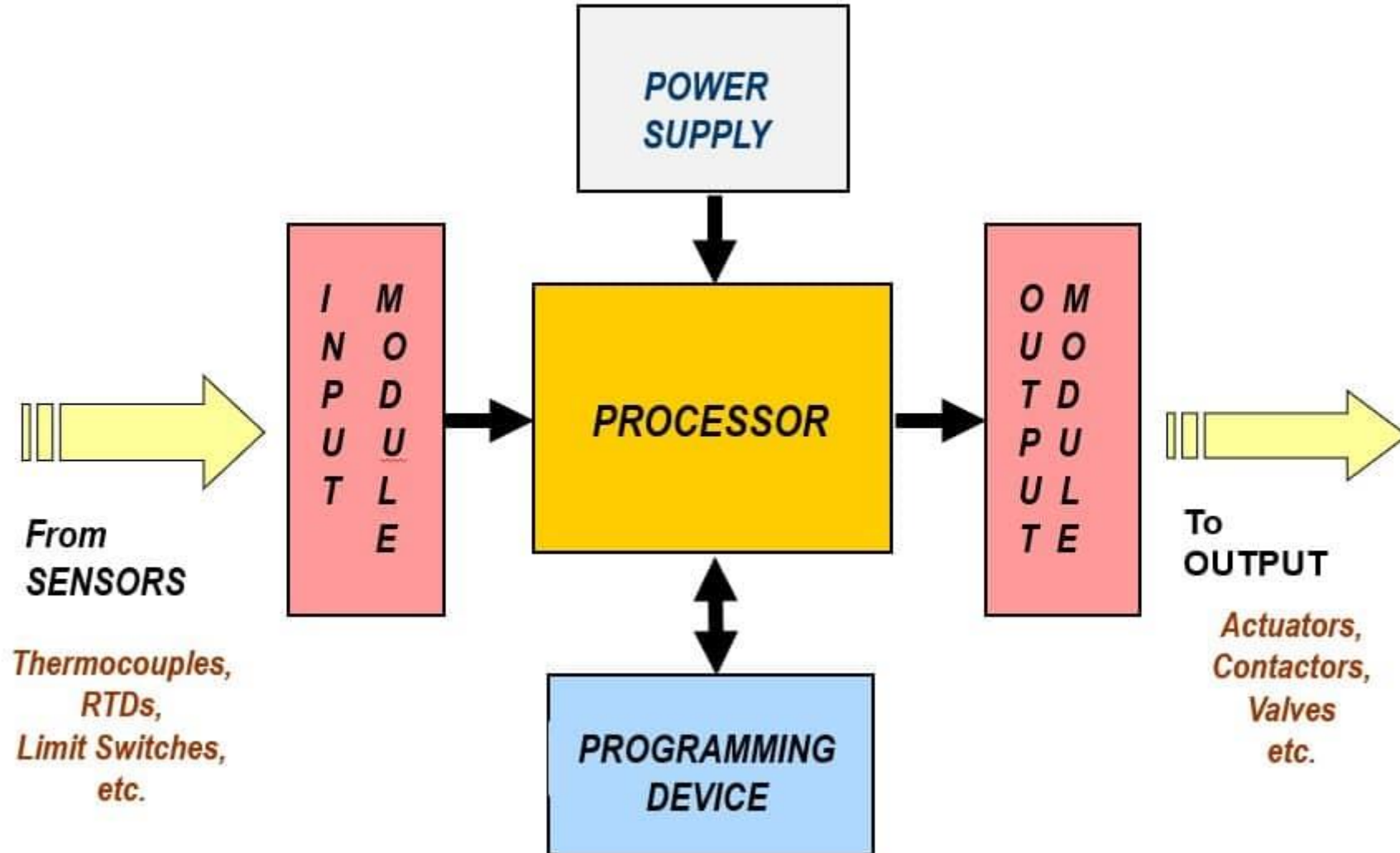
- A mechanical solution could involve a rotating cam
  - The cam would be rotated at a constant rate and the cam follower used to actuate a switch
  - The length of time for which the switch is closed depending on the shape of the cam

# PLC System

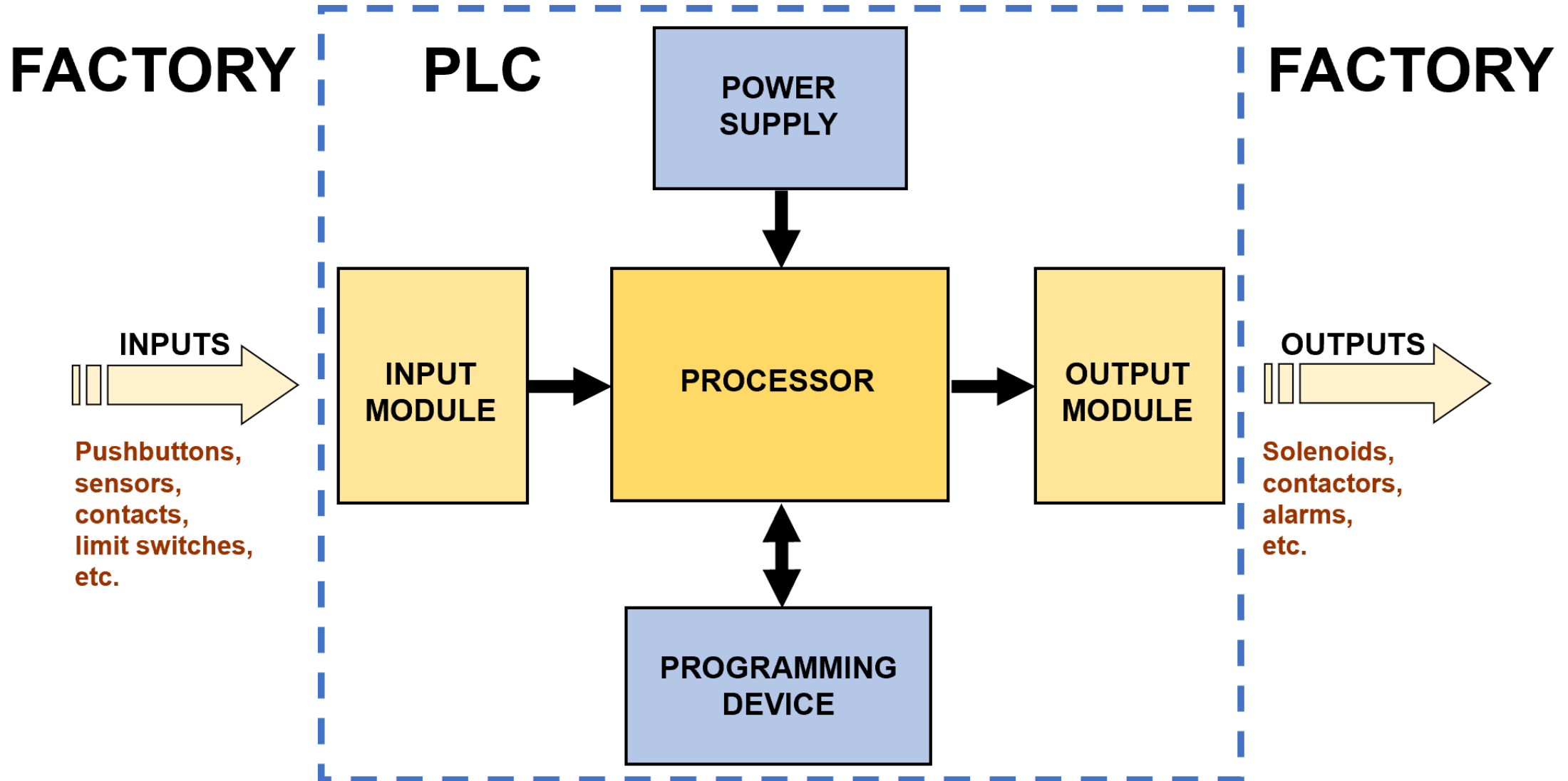
- PLCs replace complex relay circuits with a more flexible and programmable solution



# PLC System



# PLC System



- PLC as an industrial-grade microcontroller
- Excel in controlling complex machinery like assembly lines and robots
- Industrial automation
- Logic Ladder programming, function block diagrams, text-based languages

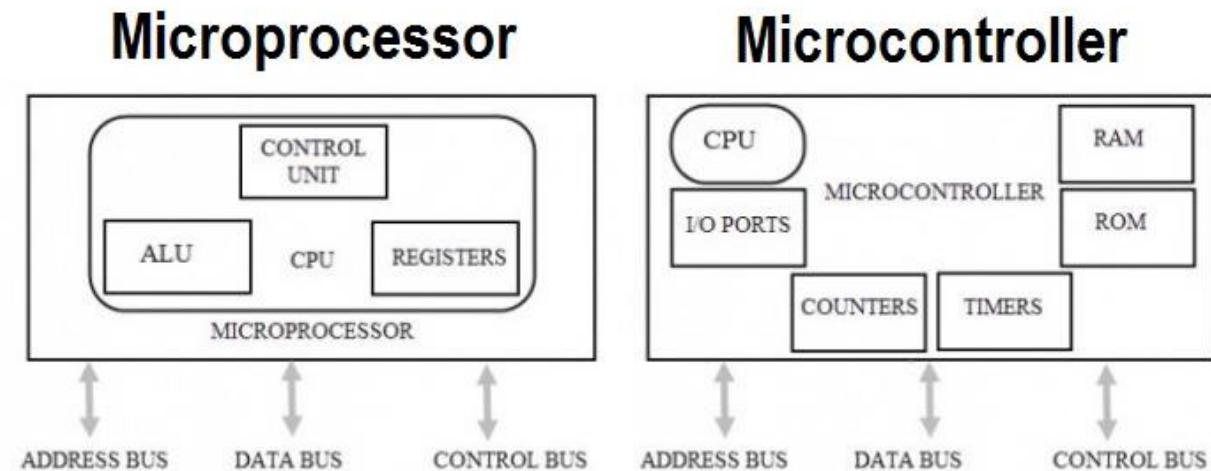
# Microprocessor and Microcontroller

## Microprocessor:

- Performs calculations and controls other parts of the system, but it needs additional components to function.
- These external components include memory (RAM and ROM) and input/output (I/O) peripherals like screens, keyboards, or sensors

## Microcontroller:

- It integrates the CPU, memory, and I/O peripherals all onto a single chip
- Microcontrollers are ideal for embedded systems, which are devices designed for a specific task, like traffic light controllers, washing machines, or even toys

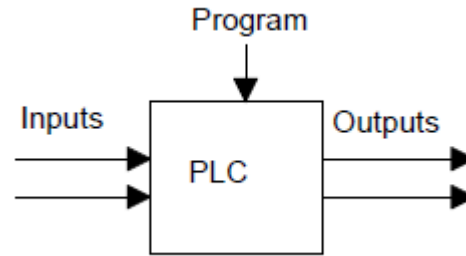




| <u>MICROPROCESSOR</u>                          | <u>MICROCONTROLLER</u>                                                   |
|------------------------------------------------|--------------------------------------------------------------------------|
| Center of a computer system.                   | Center of embedded system.                                               |
| Memory and I/O components are external to it.  | Memory and I/O components are internal to it.                            |
| Large Circuit                                  | Smaller Circuit                                                          |
| Not compatible with compact systems            | Compatible with compact systems.                                         |
| Higher cost                                    | Lower Cost                                                               |
| High Power Consumption                         | Low Power Consumption                                                    |
| Mostly don't have power features               | Mostly have power features.                                              |
| Mainly present in personal computers.          | Mainly present in washing machines, music players, and embedded systems. |
| Less number of registers.                      | More number of registers.                                                |
| Follows Von Neumann model                      | Follows Harvard architecture                                             |
| Made on a silicon-based integrated chip.       | Byproduct microprocessors and peripherals.                               |
| RAM, ROM, and other peripherals are absent.    | RAM, ROM, and other peripherals are present.                             |
| Has an external bus to interface with devices. | Uses an internal controlling bus for communication.                      |
| Has a high speed.                              | Speed depends on the architecture.                                       |
| Ideal for general purpose to handle more data. | Ideal for the specific applications.                                     |
| Complex and Expensive                          | Simple and affordable                                                    |
| Requires more instructions                     | Requires less instructions                                               |

# The programmable logic controller

Special form of *microprocessor-based controller* that uses a *programmable memory* to store instructions and to *implement functions* such as logic, sequencing, timing, counting and arithmetic *in order to control machines and processes*

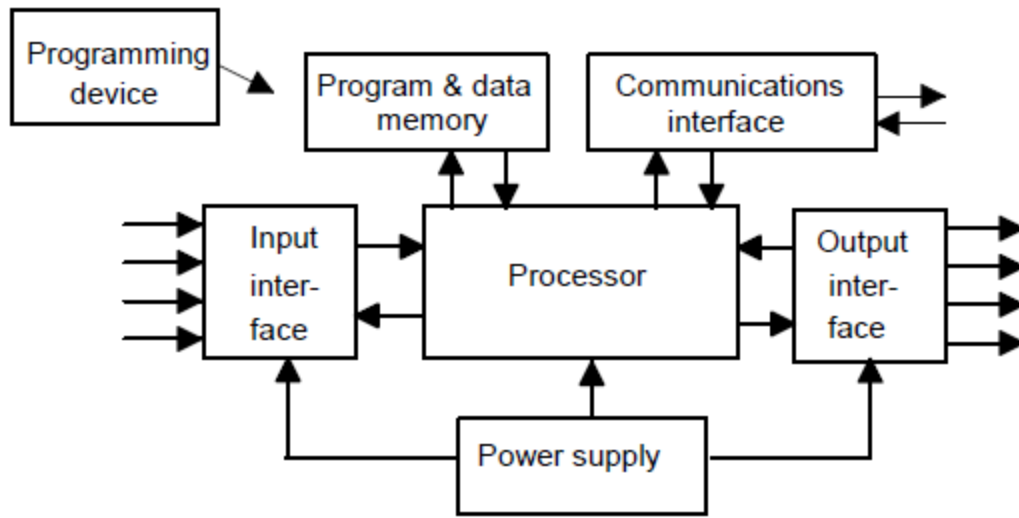


- Input devices, e.g. sensors, switches and output devices in the system being controlled, e.g. motors, valves etc., are connected to the PLC.
- The operator then enters a sequence of instructions, i.e. a program, into the memory of the PLC.
- The controller then monitors the inputs and outputs according to this program and carries out the control rules for which it has been programmed

# Features of PLC

- Same basic controller can be used with a wide range of control systems.
- To modify a control system and the rules that are to be used, all that is necessary is for an operator to input different set of instructions.
- The PLCs are flexible and cost effective
- PLCs are optimized for control tasks and the industrial environment.
- Rugged and designed to withstand vibrations, temperature, humidity and noise.
- Easily programmed and easy understand programming language - Logic Ladder programming, function block diagrams, text-based languages

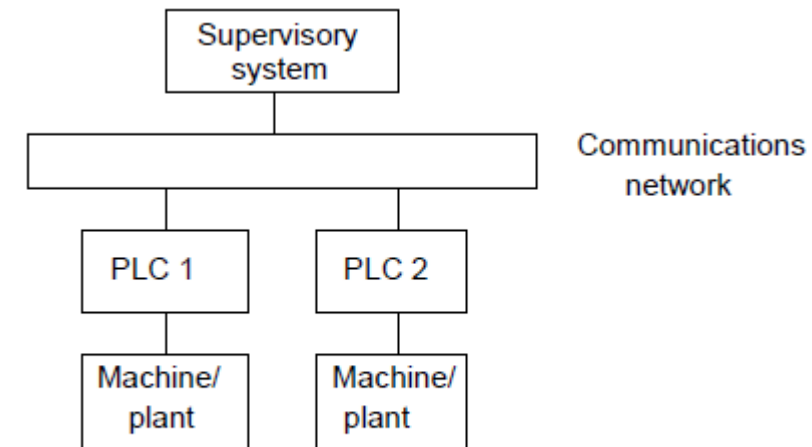
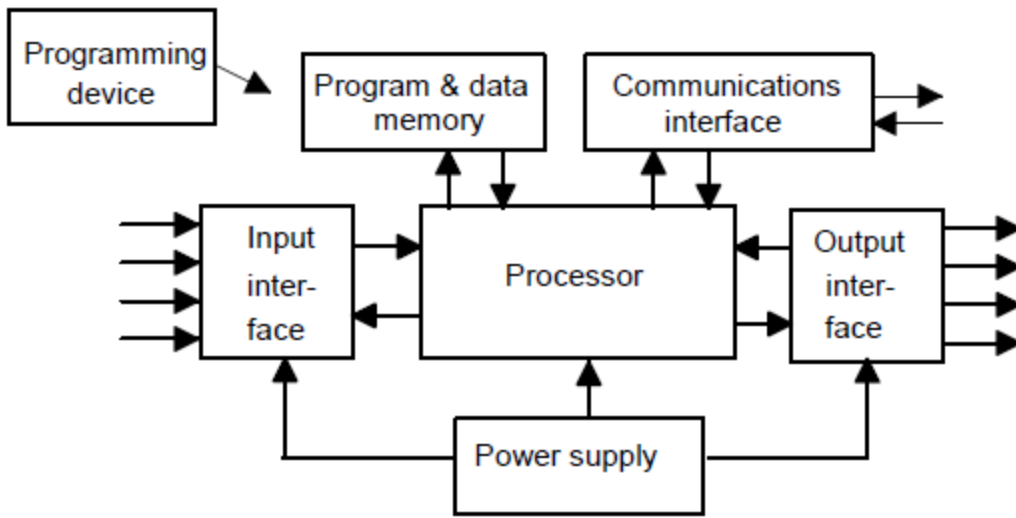
# Hardware - PLC



- The *processor unit* or *central processing unit (CPU)* is the unit containing the microprocessor and this interprets the input signals and carries out the control actions, according to the program stored in its memory, communicating the decisions as action signals to the outputs.
- The *power supply unit* is needed to convert the mains a.c. voltage to the low d.c. voltage (5 V) necessary for the processor and the circuits in the input and output interface modules.
- The *programming device* is used to enter the required program into the memory of the processor. The program is developed in the device and then transferred to the memory unit of the PLC.
- The *memory unit* is where the program is stored that is to be used for the control actions to be exercised by the microprocessor and data stored from the input for processing and for the output for outputting.

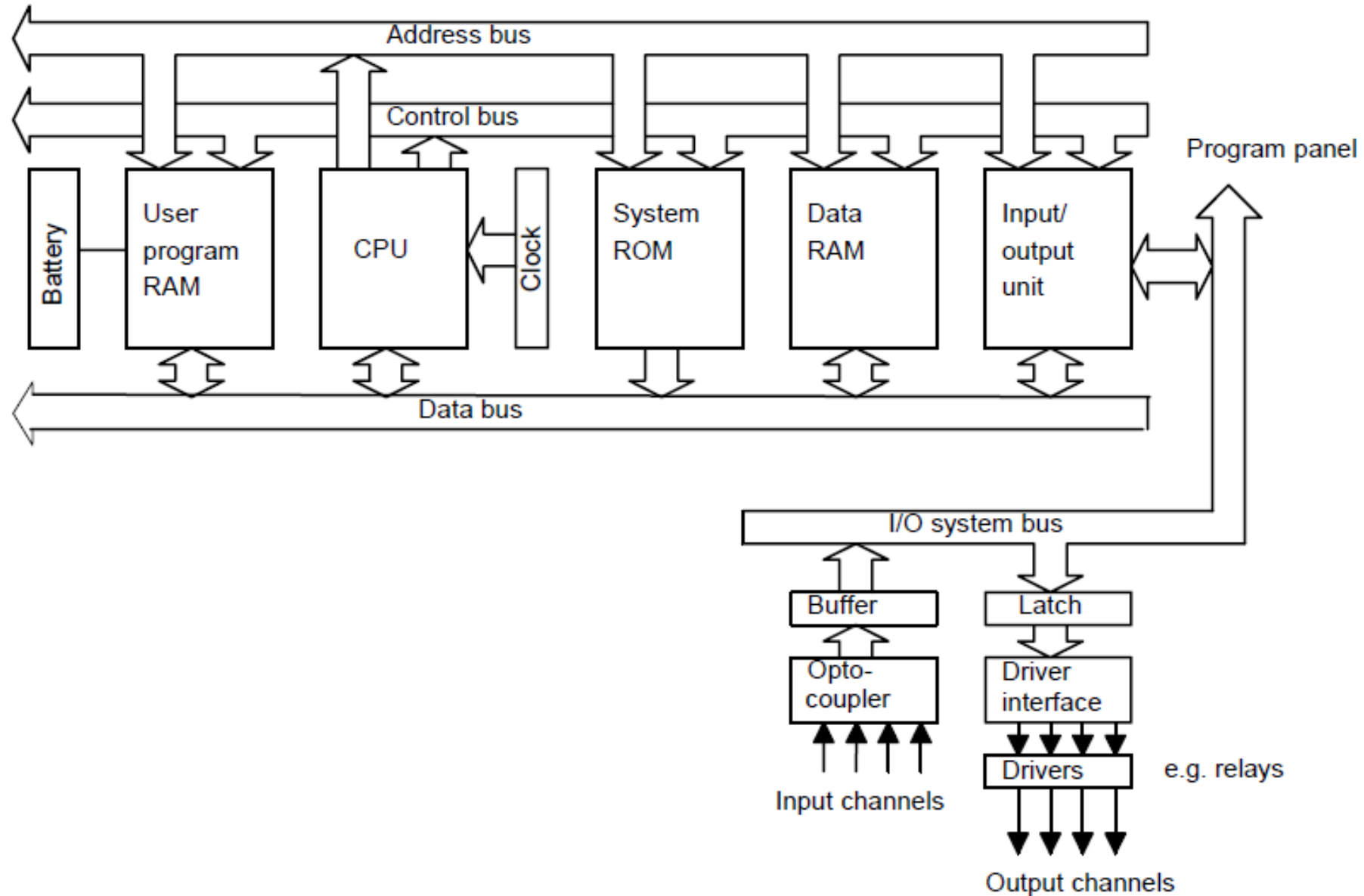
# Hardware - PLC

- The *input and output sections* are where the processor receives information from external devices and communicates information to external devices.
- The *communications interface* is used to receive and transmit data on communication networks from or to other remote PLCs.
  - The purpose is that device verification, data acquisition, synchronisation between user applications and connection management.

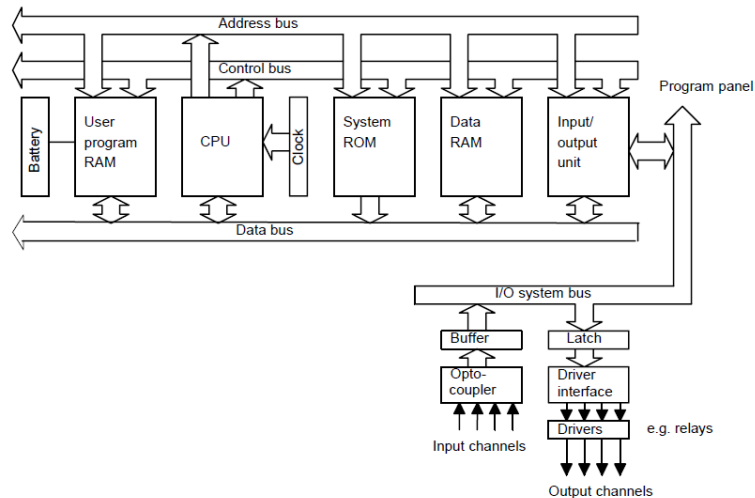


*Basic communications model*

# Internal architecture



# Internal architecture

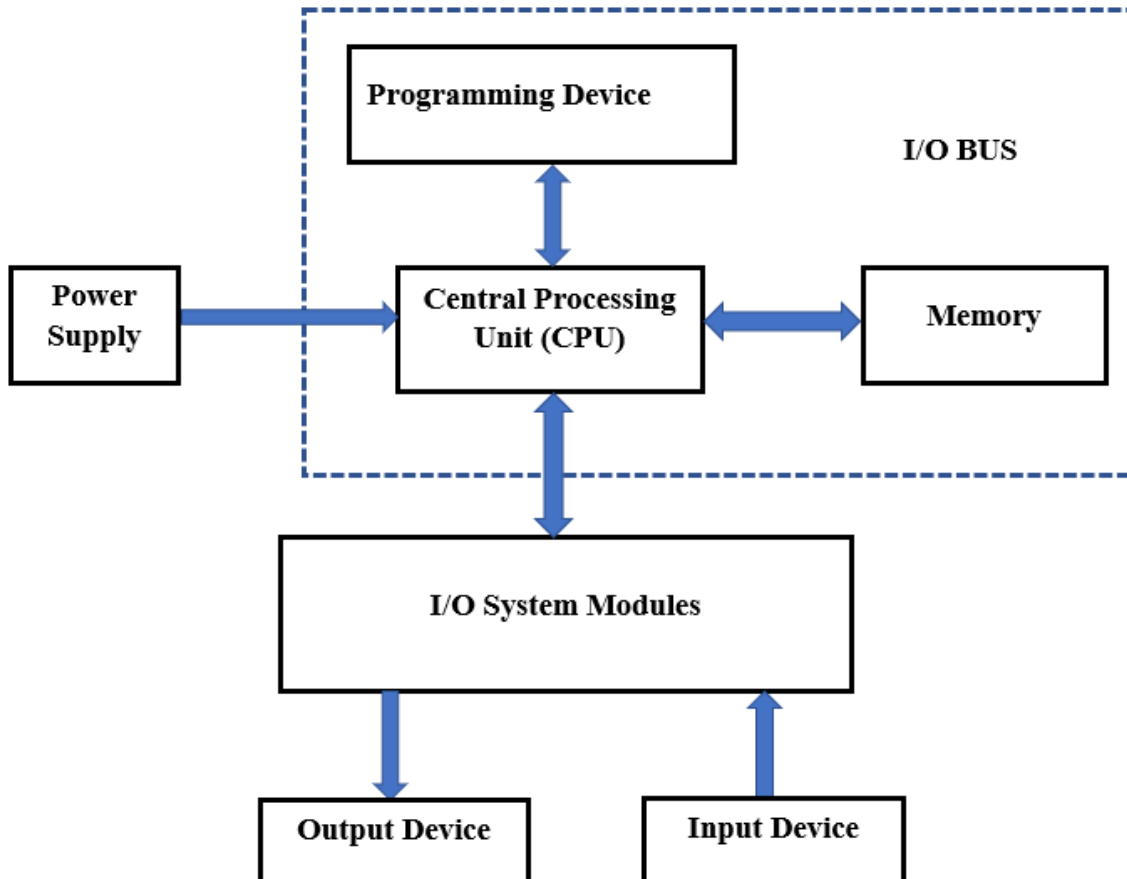


- PLC consists of a central processing unit (CPU) containing the system microprocessor, memory, and input/output circuitry.
- The CPU controls and processes all the operations within the PLC
- It is supplied with a clock with a frequency of typically between 1 and 8 MHz, this frequency determines the operating speed of the PLC
- The information within the PLC is carried by means of digital signals.
- ***The internal paths along which digital signals flow are called buses***
- The CPU uses the **data bus** for sending data between the constituent elements
- The **address bus** to send the addresses of locations for accessing stored data
- The **control bus** carries the signals used by the CPU for control
- The **system bus** is used for communications between the input/output unit

# PLC



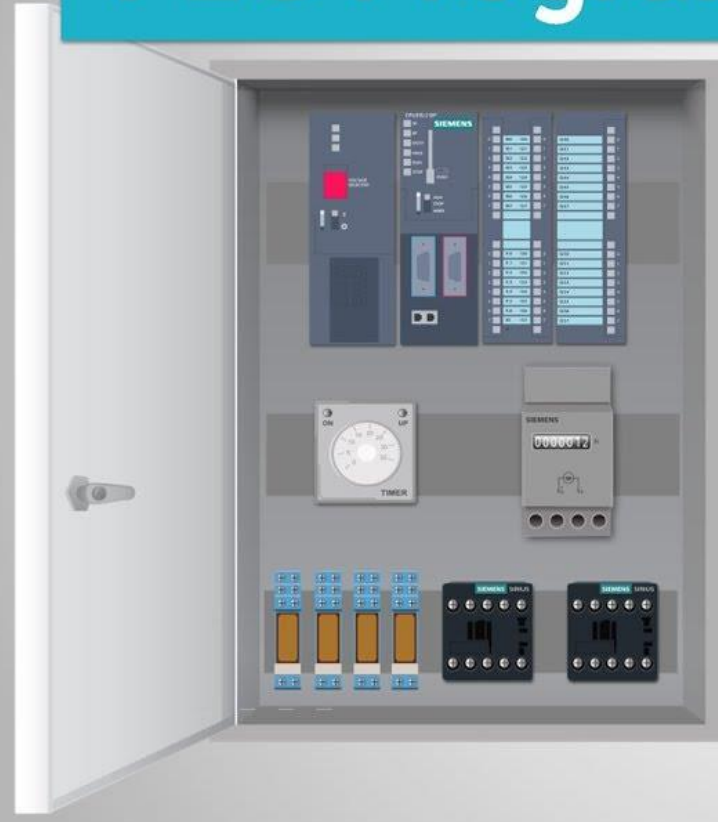
## PLC Programmable Logic Controller



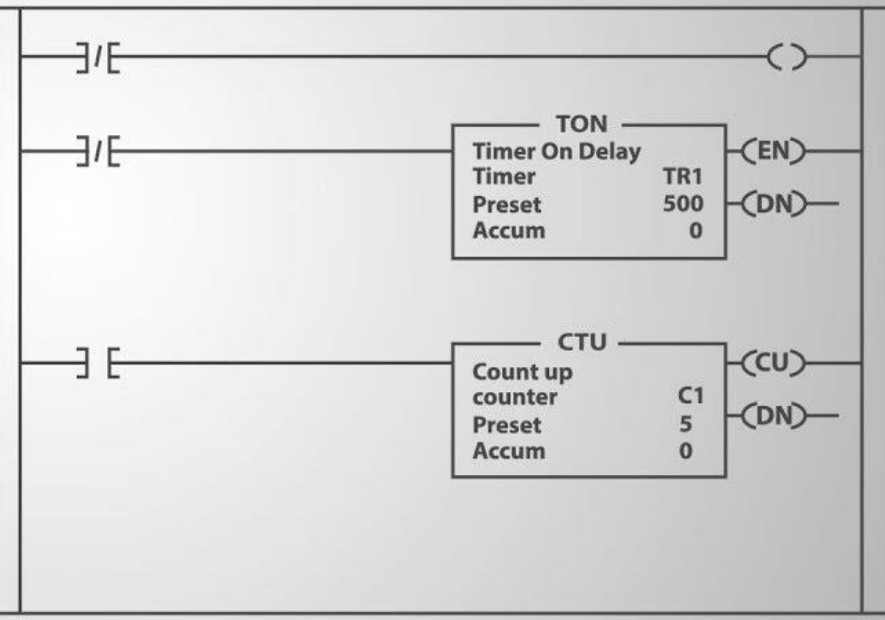


# PLC Programming Language

## PLC Programming Languages



Comments:



# PLCs

- PLCs can be used by engineers without any great knowledge of programming
- Logic ladder programming can be converted into machine code by some software for use by the PLC microprocessor
- International standard has been adopted for ladder programming in 1993, is IEC 1131-3 (International Electrotechnical Commission)
- The IEC 1131-3 programming languages are
  - Ladder diagrams (LAD)
  - Instruction list (IL)
  - Sequential function charts (SFC)
  - Structured text (ST)
  - Function block diagrams (FBD)

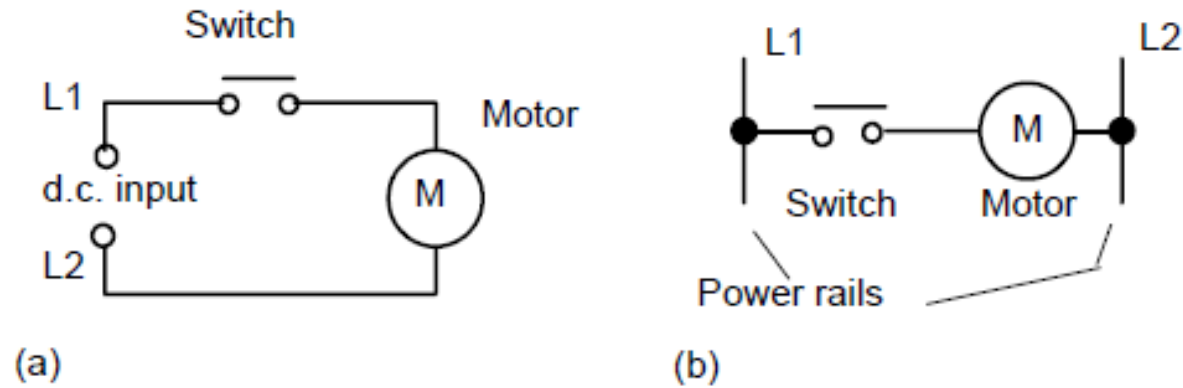
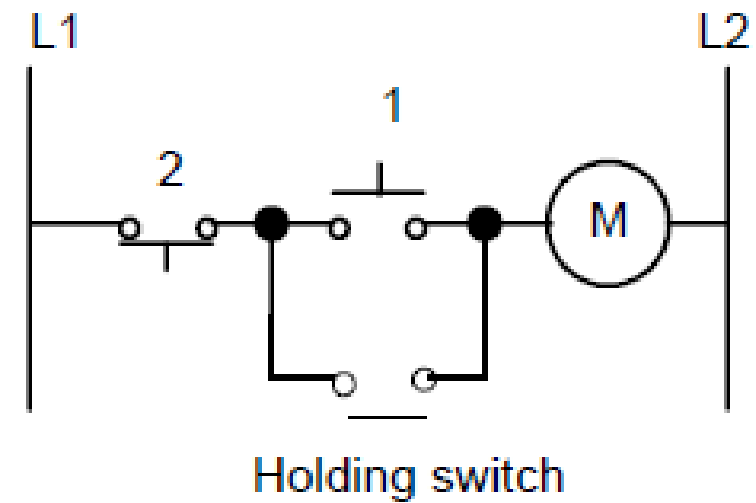


Figure 5.1 *Ways of drawing the same electrical circuit*

- The diagram shows the circuit for switching on or off an electric motor.
- We can redraw this diagram in a different way, using two vertical lines to represent the input power rails and stringing the rest of the circuit between them



- Ladder diagram for a circuit that is used to start and stop a motor using push buttons.
- In the normal state, push button 1 is open and push button 2 closed.
- When button 1 is pressed, the motor circuit is completed and the motor starts.
- Also, the holding contacts wired in parallel with the motor close and remain closed as long as the motor is running.
- Thus when the push button 1 is released, the holding contacts maintain the circuit and hence the power to the motor.
- To stop the motor, button 2 is pressed, disconnects the power to the motor and the holding contacts open
- Motor is started by pressing button 1 and stopped by pressing button 2

# PLC ladder programming

- Writing a program is then equivalent to drawing a switching circuit.
  - The ladder diagram consists of two vertical lines representing the power rails
  - Circuits are connected as horizontal lines, i.e. the rungs of the ladder, between these two verticals

In drawing a ladder diagram, certain conventions are adopted

1. The vertical lines of the diagram represent the power rails between which circuits are connected. The power flow is taken to be from the left-hand vertical across a rung.
2. Each rung on the ladder defines one operation in the control process
3. A ladder diagram is read from left to right and from top to bottom
  - When the PLC is in its run mode, it goes through the entire ladder program to the end termed a *cycle*.
  - The end rung might be indicated by a block with the word END

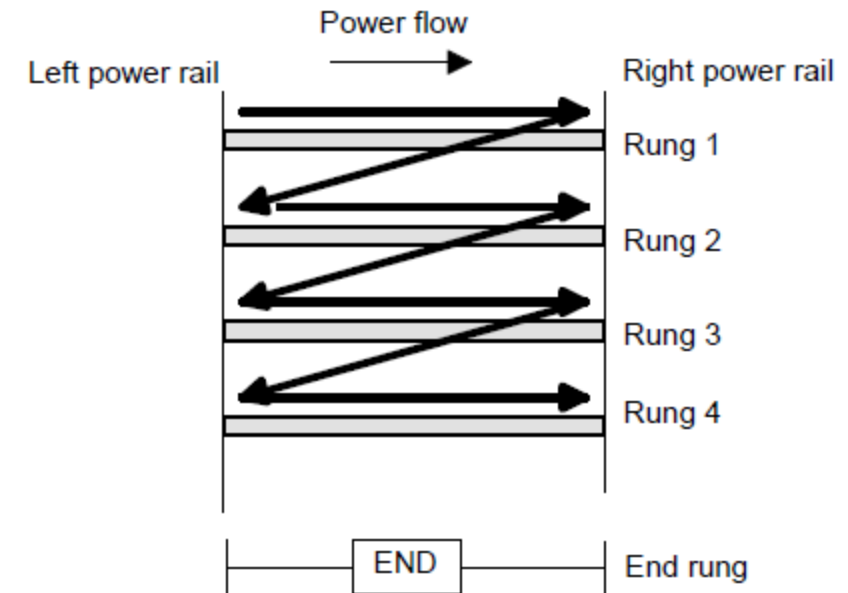


Figure 5.3 Scanning the ladder program

# PLC ladder programming

4. Each rung must start with an input or inputs and must end with at least one output
  - The term input is used for a control action of a PLC, ex: closing a switch
  - The term output is used for a device connected to the output of a PLC, ex. a motor
5. Electrical devices are shown in their normal condition
  - NO switch normally open until some object closes it
  - NC switch normally closed until some object opens it
6. A particular device can appear in more than one rung of a ladder.

Ex: a relay which switches on one or more devices, same letters and/or numbers are used to label the device in each situation
7. The inputs and outputs are all identified by their *addresses*, the notation used depending on the PLC manufacturer

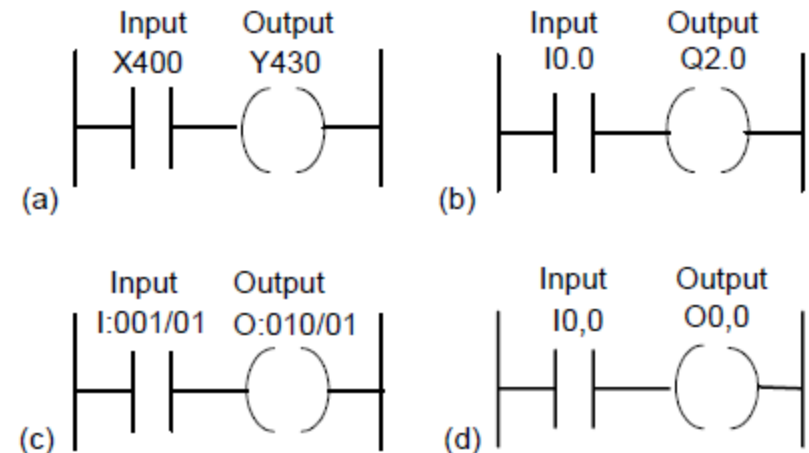


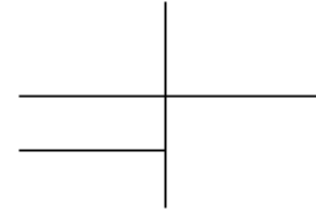
Figure 5.6 Notation: (a) Mitsubishi, (b) Siemens, (c) Allen-Bradley, (d) Telemecanique

# Basic symbols

A horizontal link along which  
power can flow



Interconnection of horizontal  
and vertical power flows



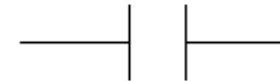
Left-hand power connection  
of a ladder rung



Right-hand power connection  
of a ladder rung



Normally open contact



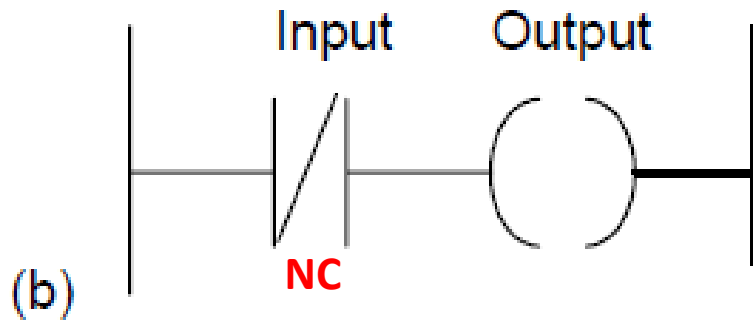
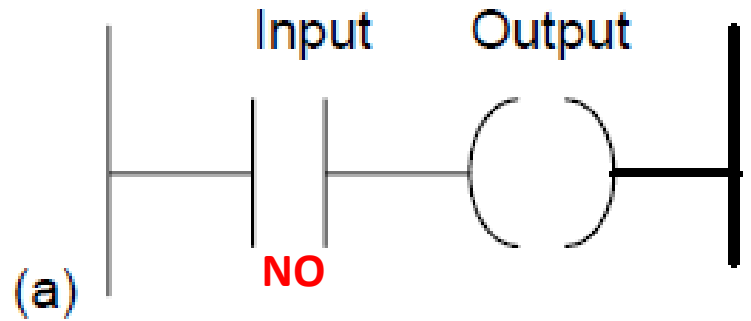
Normally closed contact



Output coil: if the power flow  
to it is on then the coil state is on



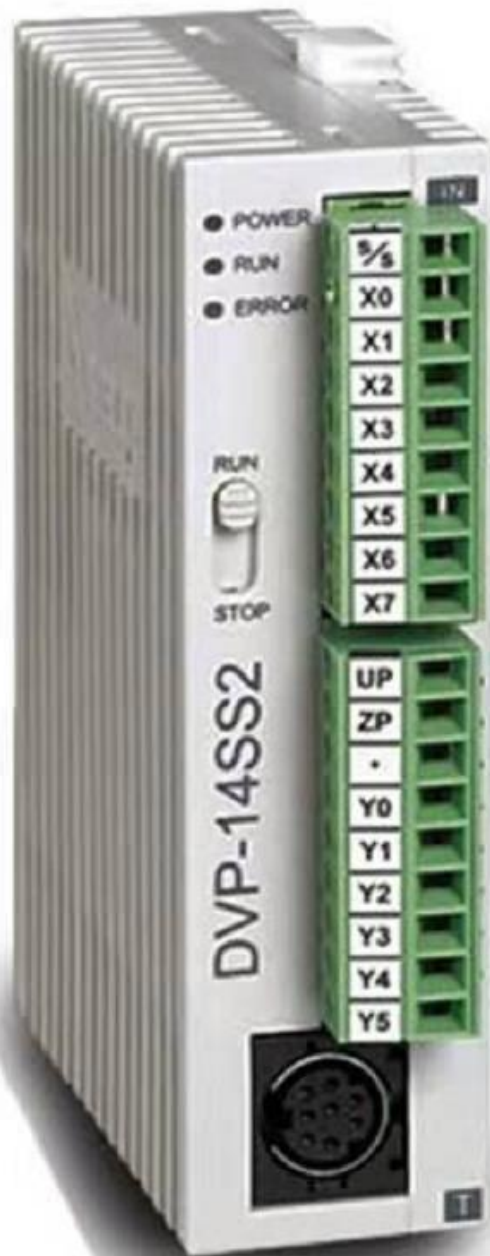
# Basic symbols - A ladder rung



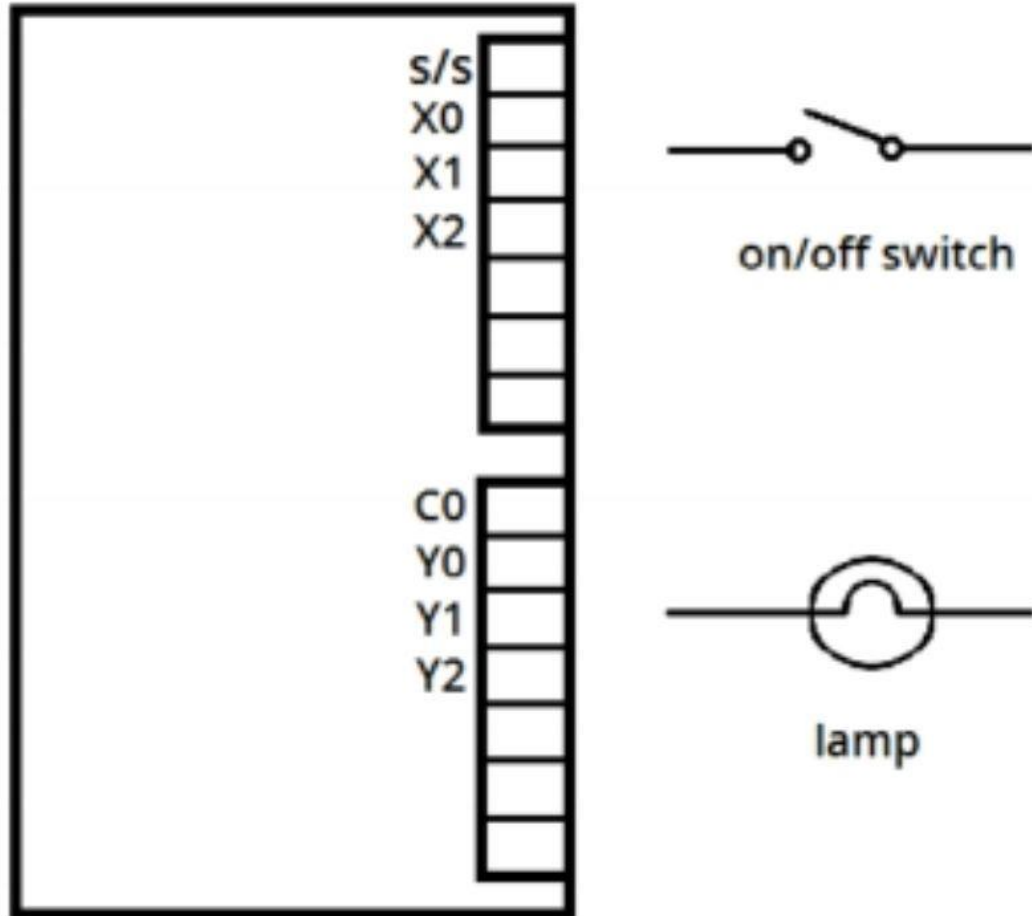
Starting with the input, we have the normally open symbol  $| |$  for the input contacts. There are no other input devices and the line terminates with the output, denoted by the symbol  $( )$ . When the switch is closed, i.e. there is an input, the output of the motor is activated. Only while there is an input to the contacts is there an output. If there had been a normally closed switch  $| / |$  with the output (Figure 5.5(b)), then there would have been an output until that switch was opened. Only while there is no input to the contacts is there an output.



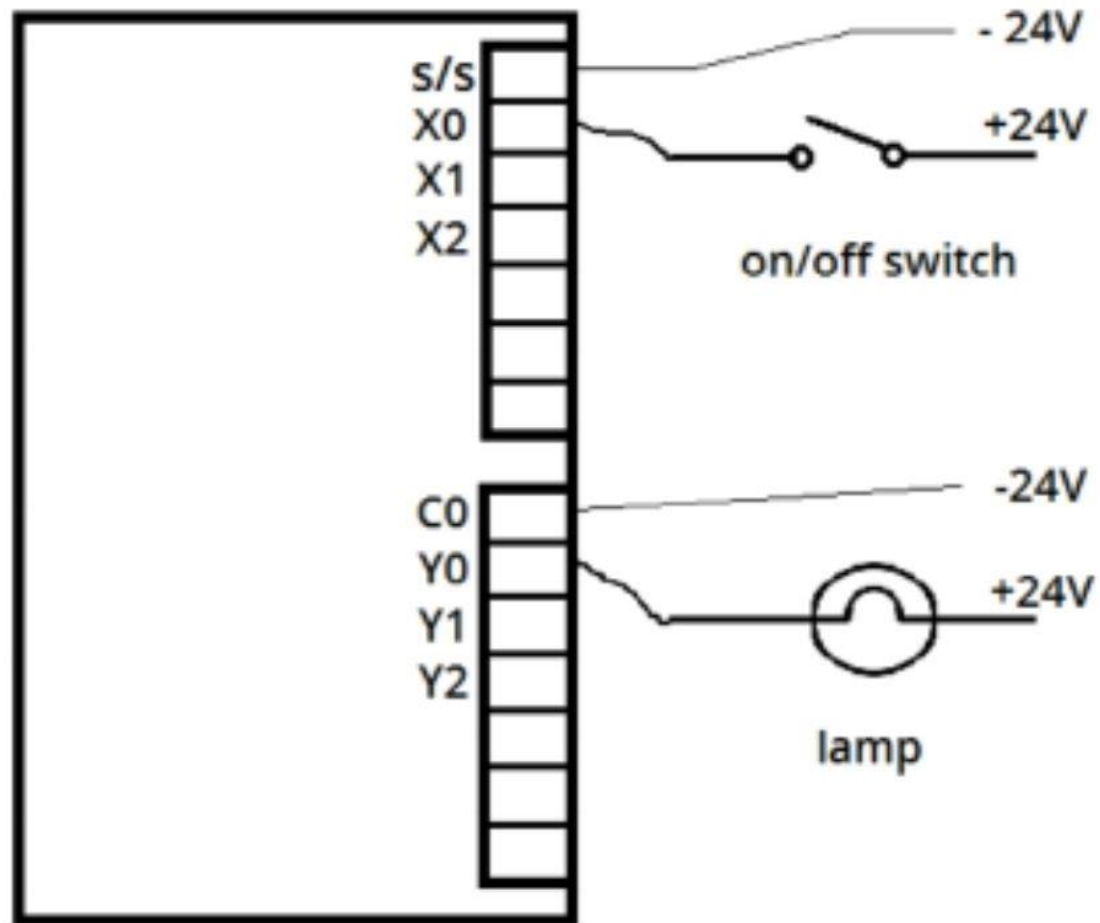
# PLC



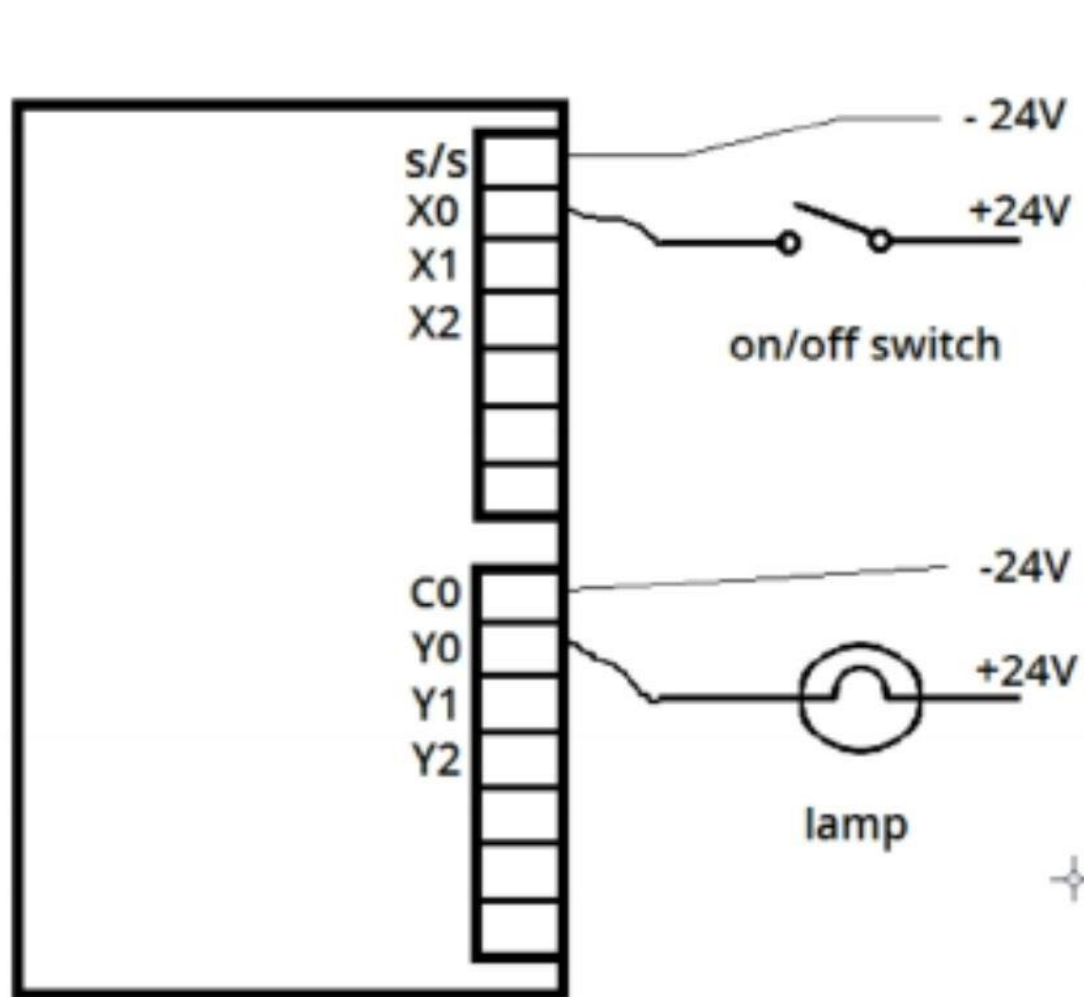
# Example Problem



# Example Problem

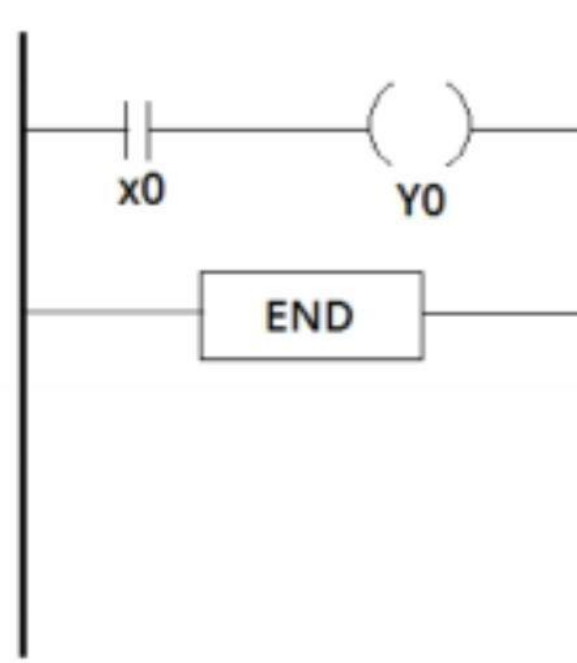


# Example Problem



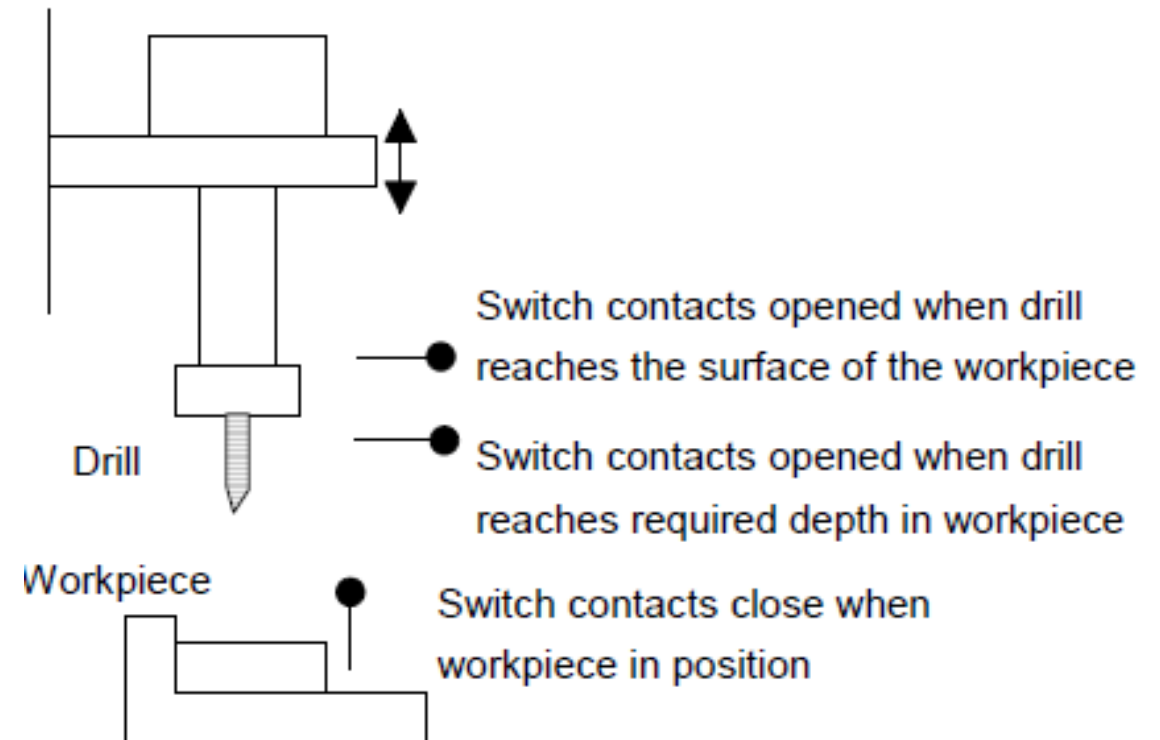
## LADDER DIAGRAM

When the switch is moved to the ON position, the lamp glows.



# Logic functions

- The condition that the drill motor is to be activated when the limit switches are activated that indicate the presence of the workpiece in position
- The situation involves the **AND logic function**, condition A and condition B having both to be realised for an output to occur



# AND

- Only when A and B are both on is there an output

| Inputs |   | Output |
|--------|---|--------|
| A      | B |        |
| 0      | 0 | 0      |
| 0      | 1 | 0      |
| 1      | 0 | 0      |
| 1      | 1 | 1      |

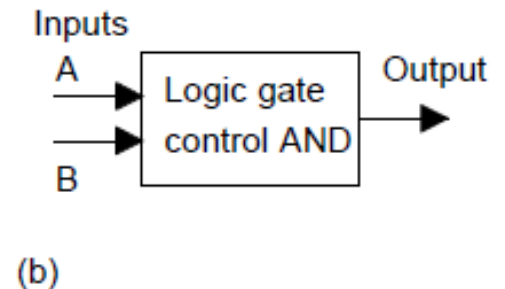
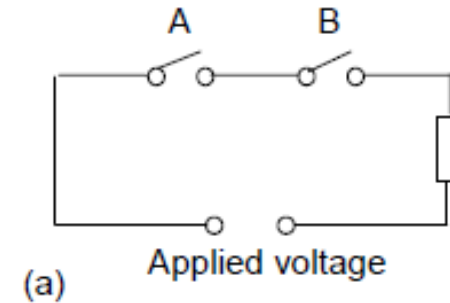
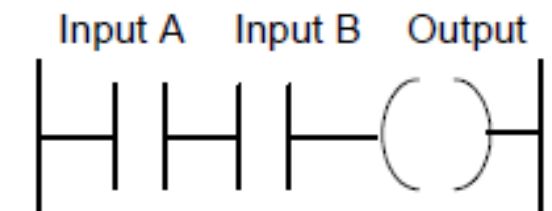


Figure 5.7 (a) AND circuit, (b) AND logic gate

- On a ladder diagram contacts in a horizontal rung, i.e. contacts logical AND operations



AND gate with a ladder diagram rung

# OR

- Input A or input B must be on for there to be an output

| Inputs |   | Output |
|--------|---|--------|
| A      | B |        |
| 0      | 0 | 0      |
| 0      | 1 | 1      |
| 1      | 0 | 1      |
| 1      | 1 | 1      |

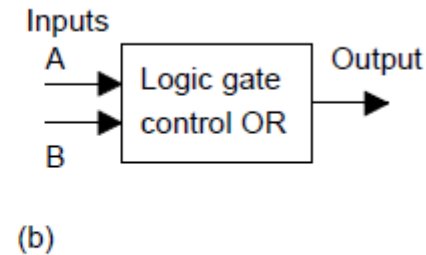
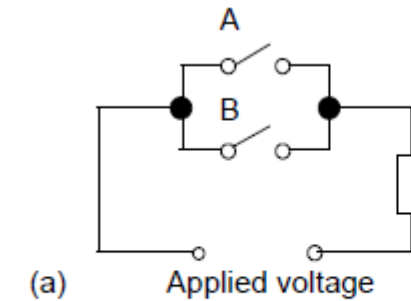
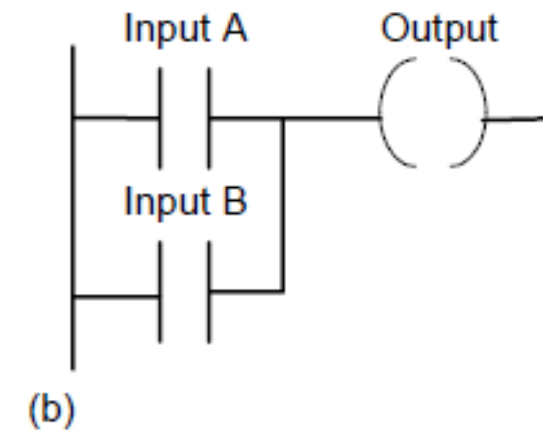
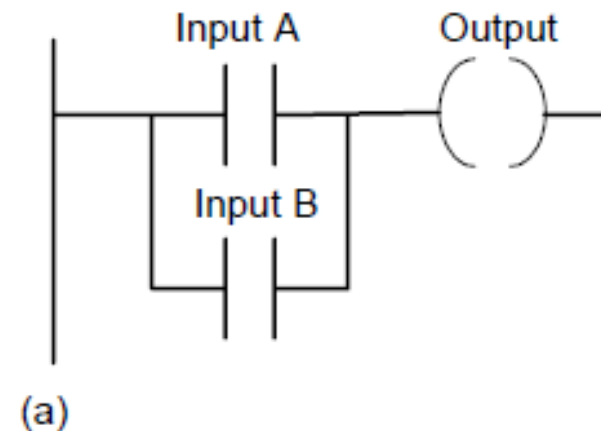


Figure 5.9 (a) OR electrical circuit, (b) OR logic gate

- Alternative paths provided by vertical paths from the main rung of a ladder diagram, i.e. paths in parallel, represent logical OR operations.*



# NOT

- There is an output when there is no input and no output when there is an input

| Input<br>A | Output |
|------------|--------|
| 0          | 1      |
| 1          | 0      |

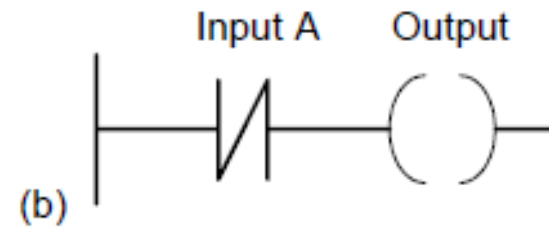
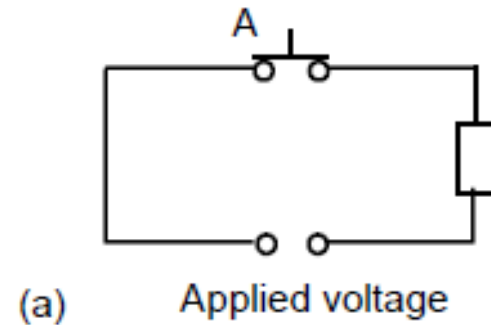


Figure 5.11 (a) NOT circuit, (b) NOT logic with a ladder rung,



# NAND

- Suppose we follow an AND gate with a NOT gate
- The consequence of having the NOT gate is to invert all the outputs from the AND gate

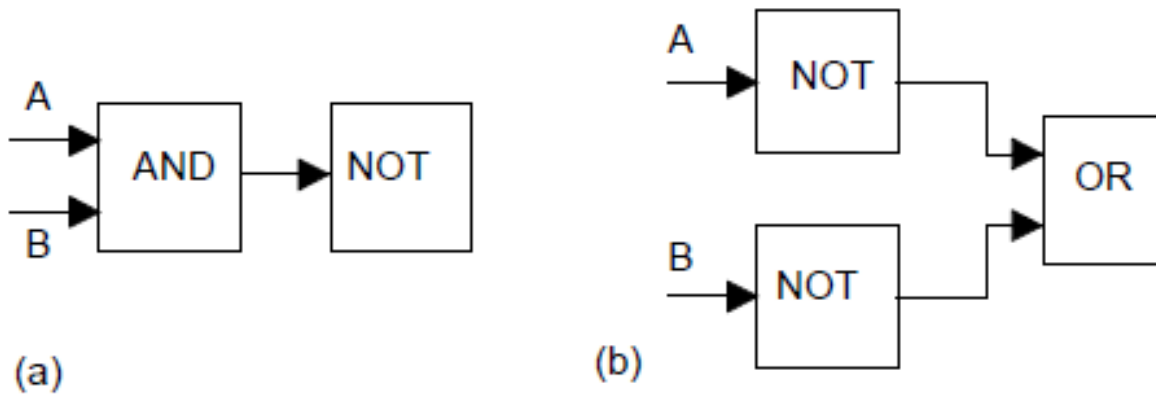


Figure 5.12 *NAND gate*

| Inputs |   | Output |
|--------|---|--------|
| A      | B |        |
| 0      | 0 | 1      |
| 0      | 1 | 1      |
| 1      | 0 | 1      |
| 1      | 1 | 0      |

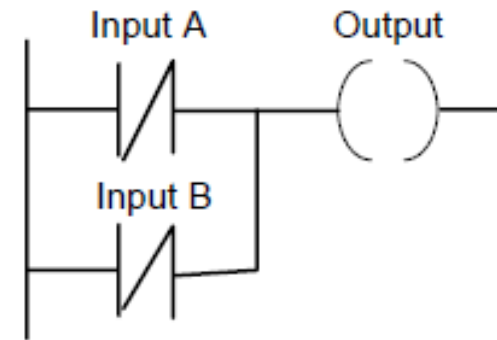


Figure 5.13 *A NAND gate*

# NOR

- Suppose we follow an OR gate by a NOT gate
- The consequence of having the NOT gate is to invert the outputs of the OR gate

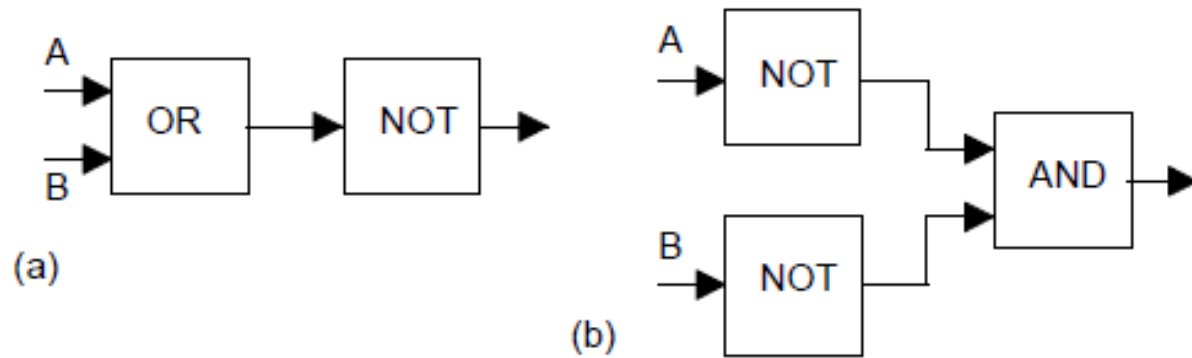


Figure 5.14 *NOR gate*

| Inputs |   | Output |
|--------|---|--------|
| A      | B |        |
| 0      | 0 | 1      |
| 0      | 1 | 0      |
| 1      | 0 | 0      |
| 1      | 1 | 0      |

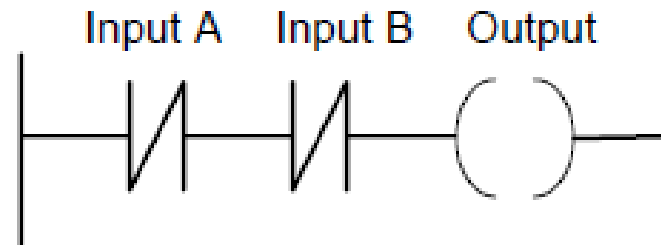


Figure 5.15 *NOR gate*

# Exclusive OR (XOR)

- The OR gate gives an output when either or both of the inputs are 1
- Sometimes there is *a need for a gate that gives an output when either of the inputs is 1 but not when both are 1*

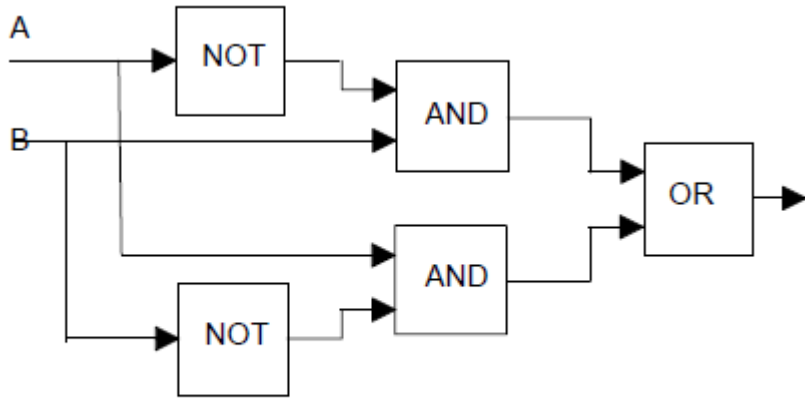


Figure 5.16 *XOR gate*

| Inputs |   | Output |
|--------|---|--------|
| A      | B |        |
| 0      | 0 | 0      |
| 0      | 1 | 1      |
| 1      | 0 | 1      |
| 1      | 1 | 0      |

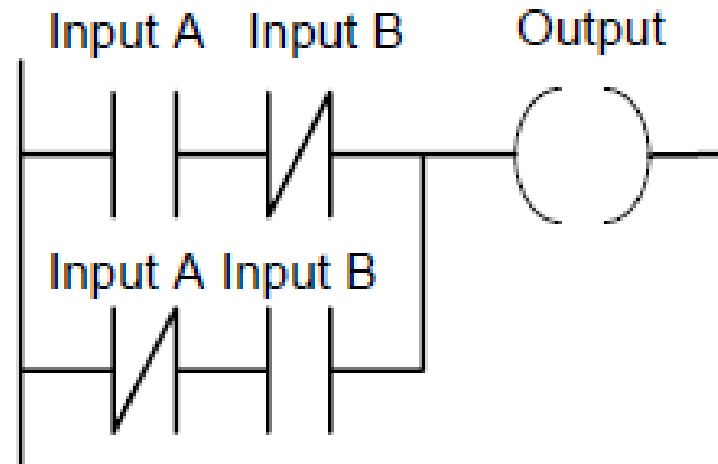


Figure 5.17 *XOR gate*

# Latching

- Consider a situations where it is necessary to hold an output energized, even when the input ceases.
- A simple example of a motor which is started by pressing a push button switch
- Though the switch contacts do not remain closed, the motor is required to continue running until a stop push button switch is pressed
- *Latching is a self-maintaining circuit in that, after being energized, it maintains that state until another input is received*

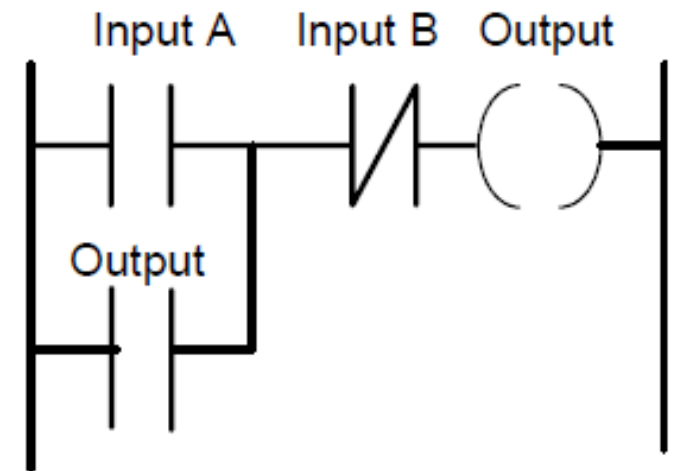
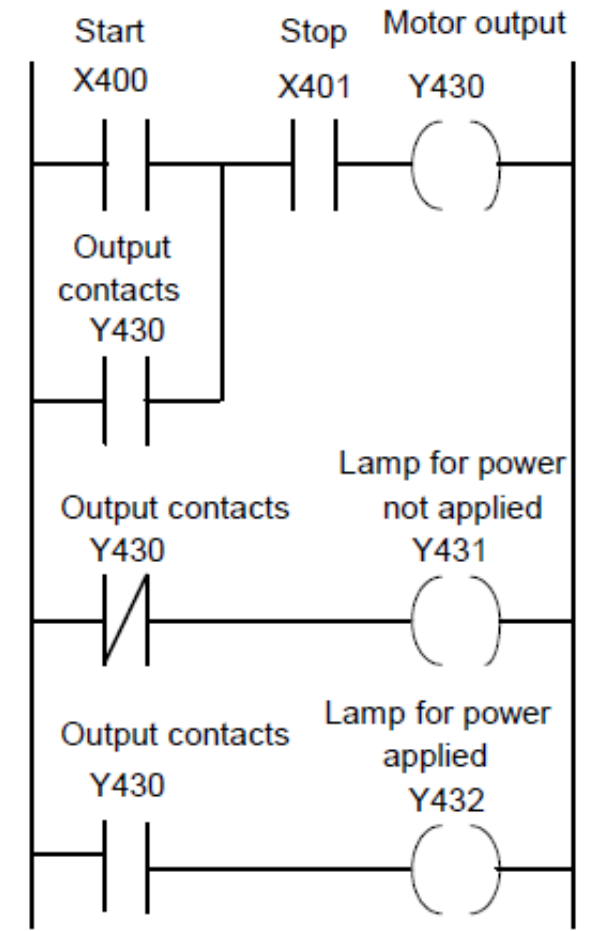


Figure 5.18 *Latched circuit*



- X401 is closed when the program is started
- When X400 is momentarily closed, Y430 is energised and its contacts close.
- This results in latching and also the switching off of Y431 and the switching on of Y432.
- To switch the motor off, X401 is pressed and opens.
- Y430 contacts open in the top rung and third rung, but close in the second rung. Thus Y431 comes on and Y432 off.

# Multiple outputs

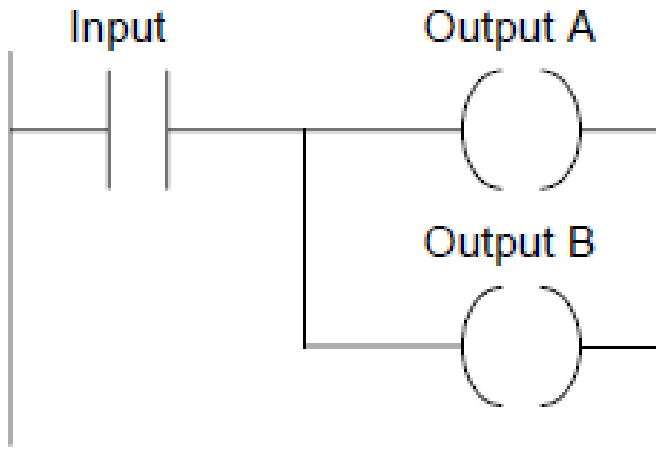
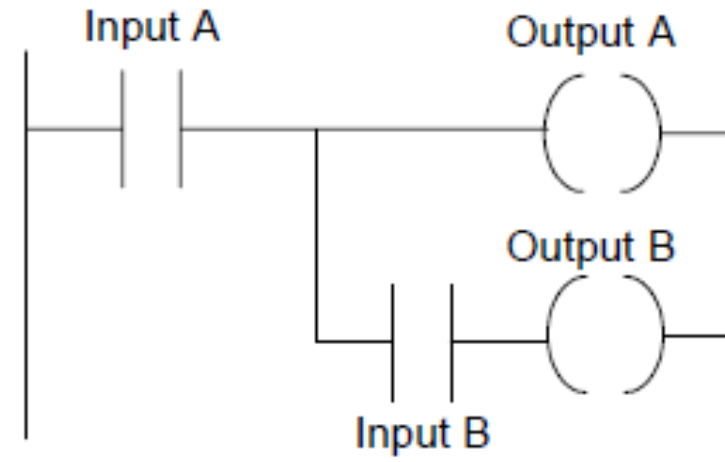
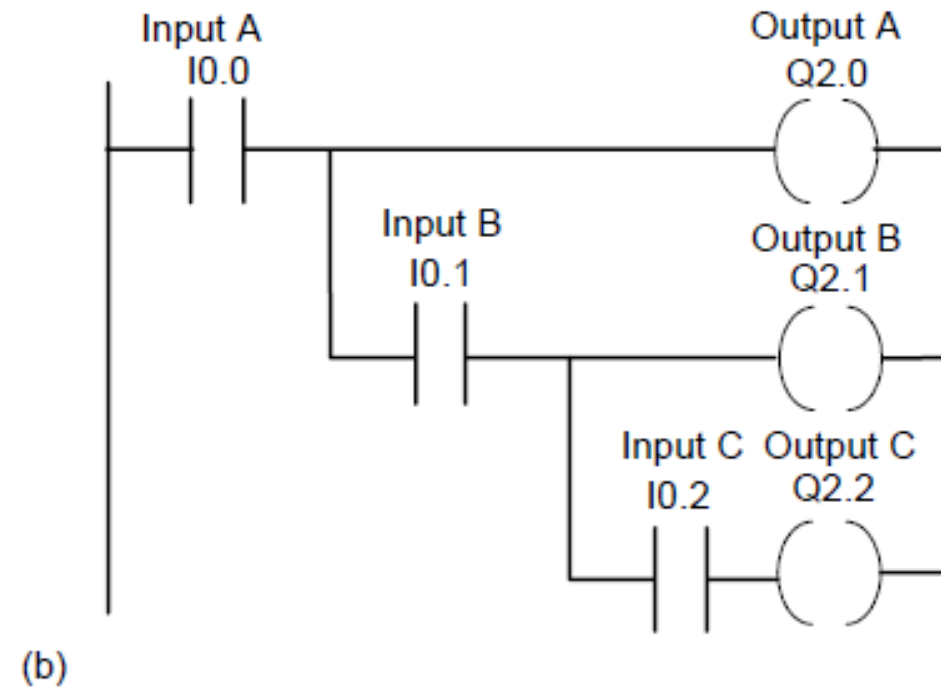
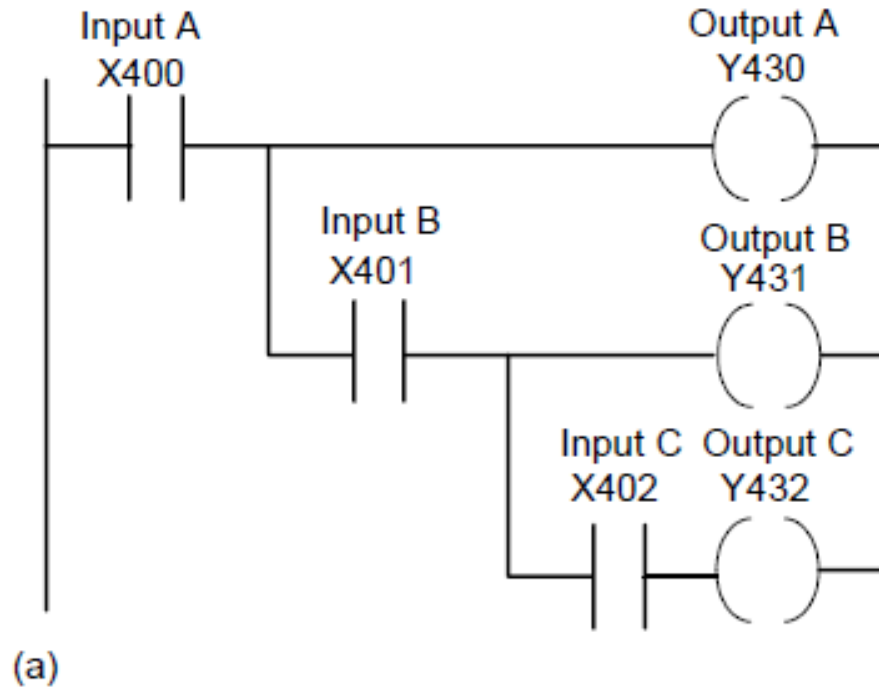


Figure 5.20 *Ladder rung with two output.*

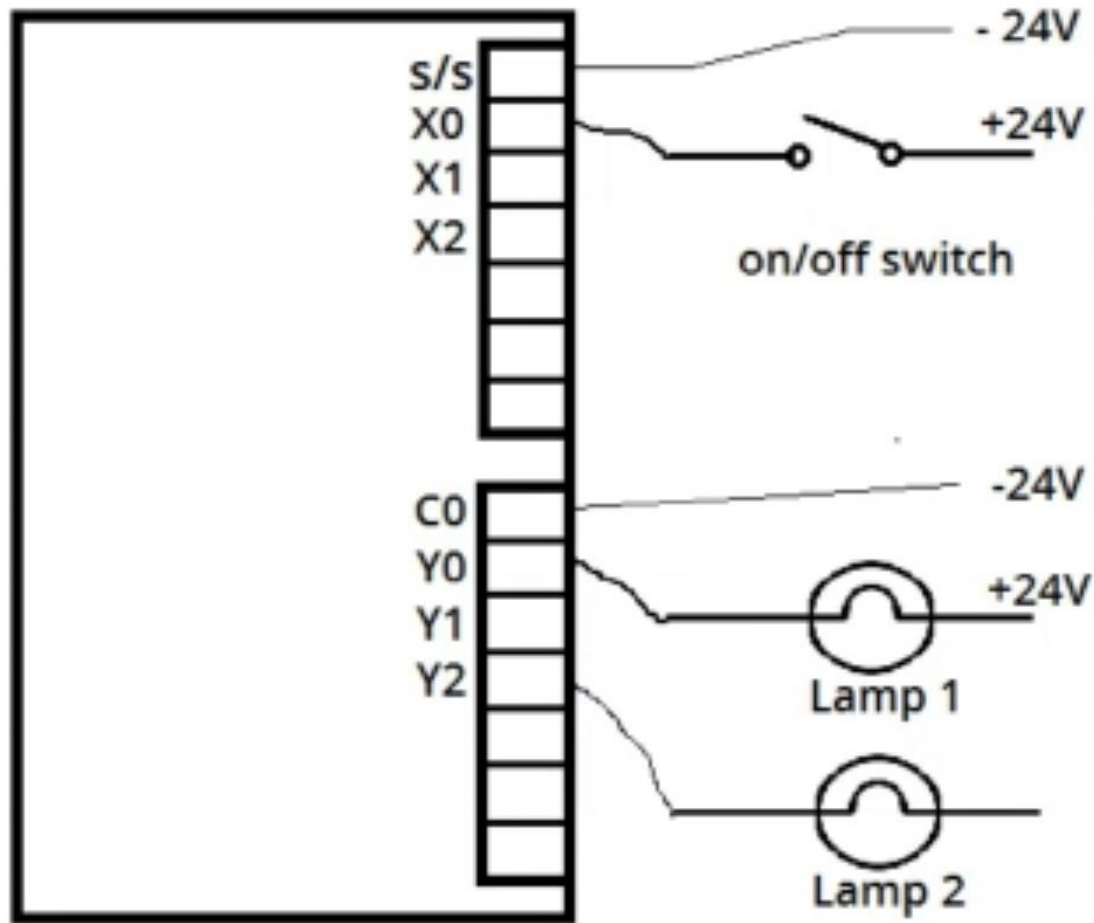


*Ladder rung with two inputs and two outputs*

# Sequenced outputs

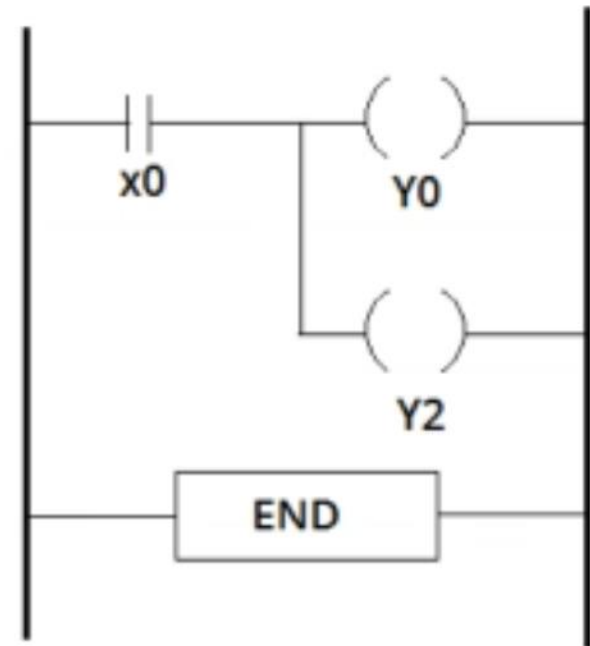


# Class room Practices



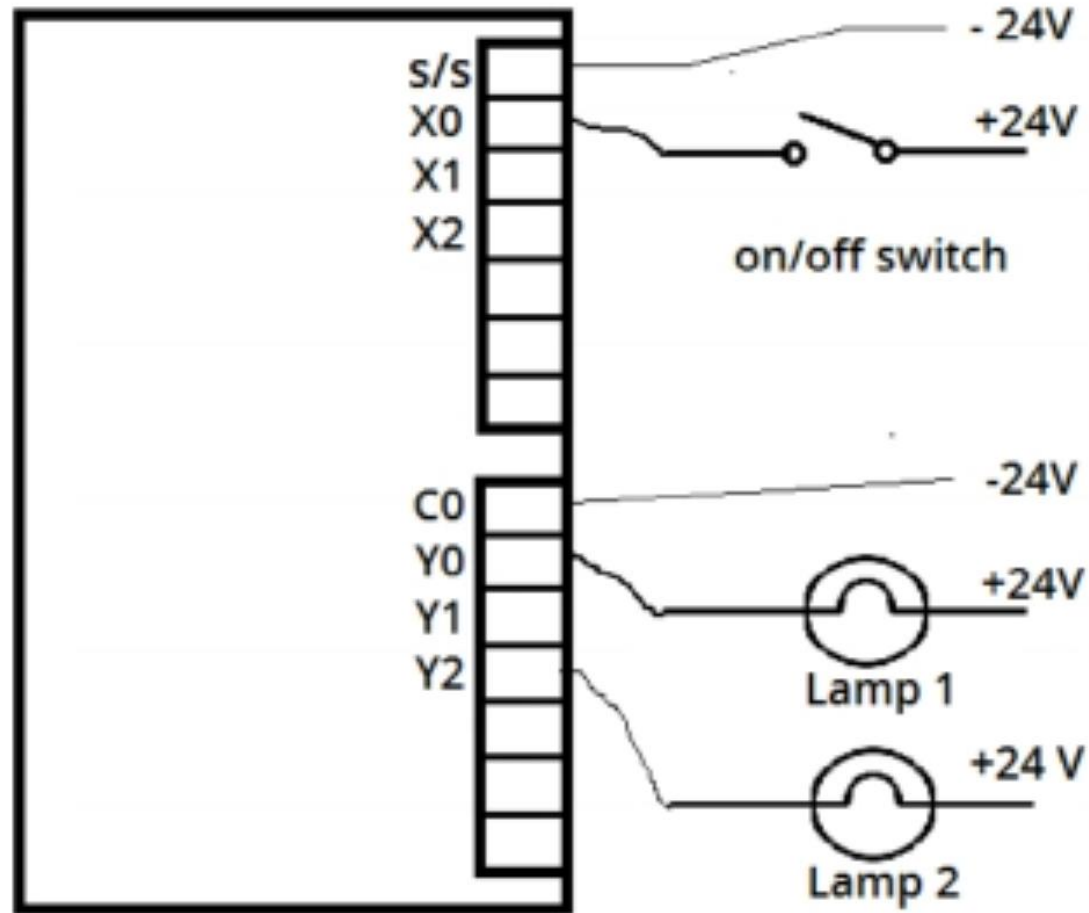
## LADDER DIAGRAM

When the switch is moved to ON position, **BOTH** lamps should glow



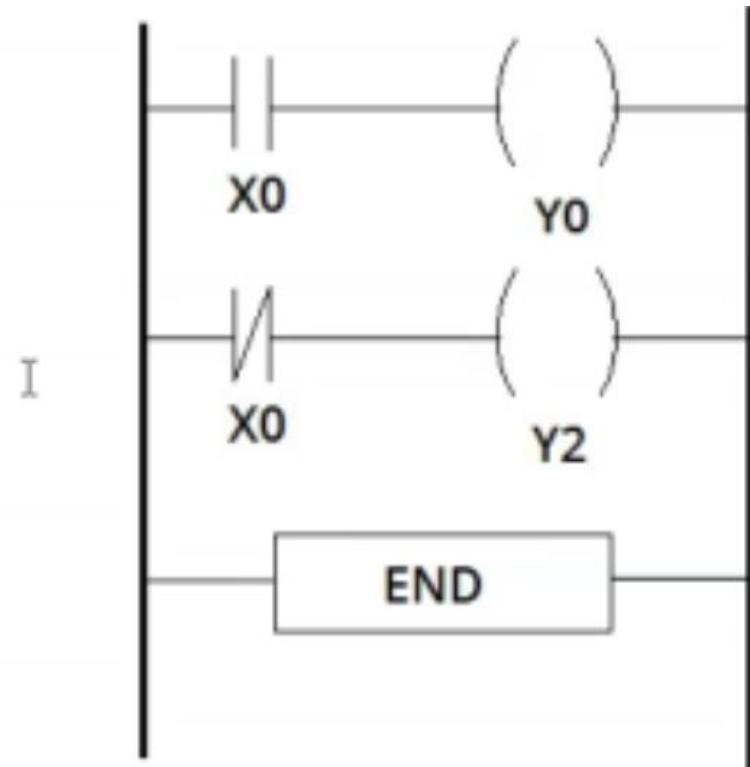


# Class room Practices



## LADDER DIAGRAM

When the switch is moved to the ON position, Lamp-1 should turn on and Lamp-2, which is initially on should turn off



# Logic gates

- United States standard
  - International standard (IEEE/ANSI)
- uses a rectangle with the logic function written inside it

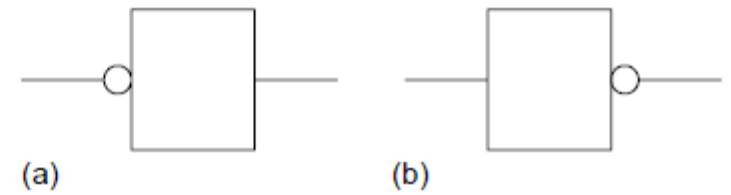
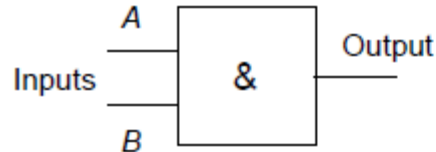
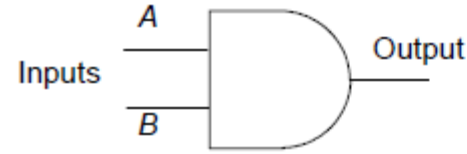
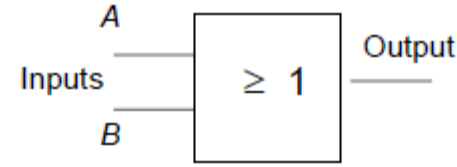
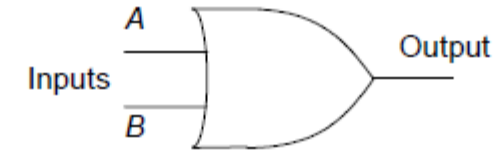


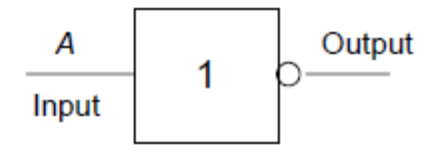
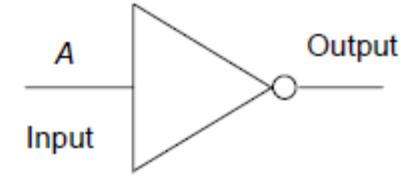
Figure 5.25 (a) Negated input, (b) negated output



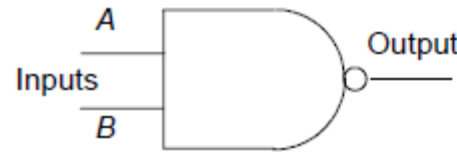
AND gate



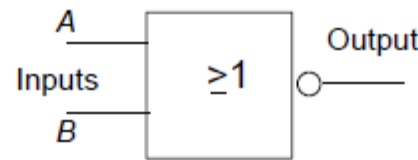
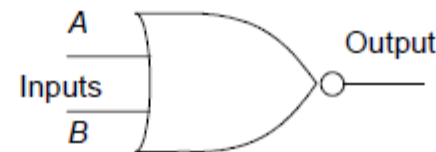
OR gate



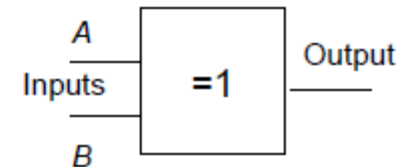
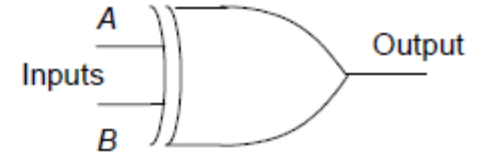
Not gate



NAND gate



NOR Gate



XOR Gate

Figure 5.26 Logic gate symbols

# Ladder diagram and equivalent functional block diagram

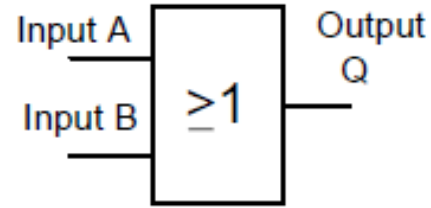
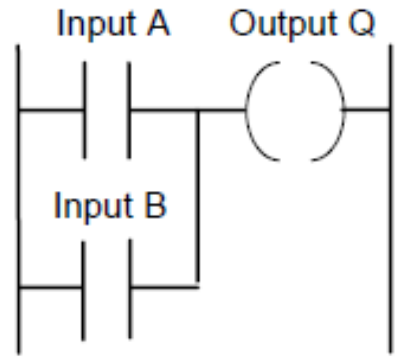


Figure 5.28 Ladder diagram and equivalent functional block diagram

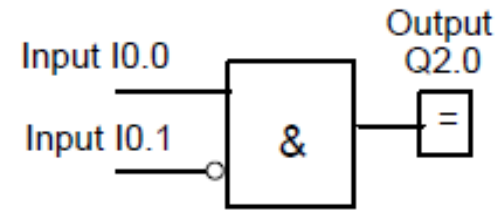
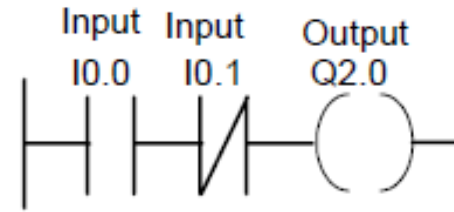


Figure 5.29 Ladder diagram and equivalent function block diagram

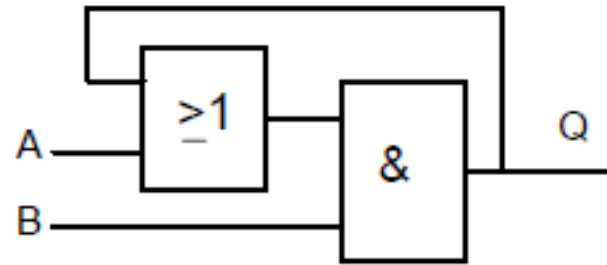
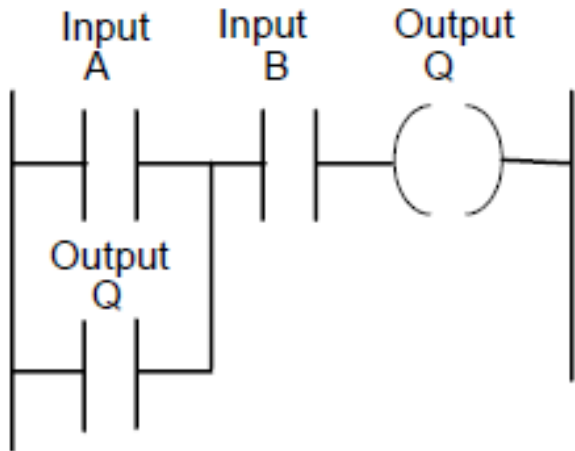
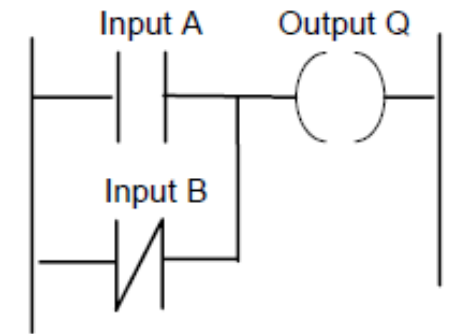
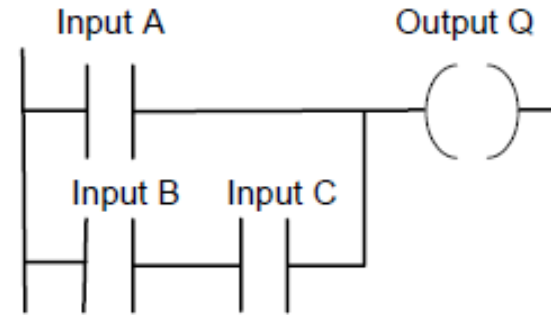


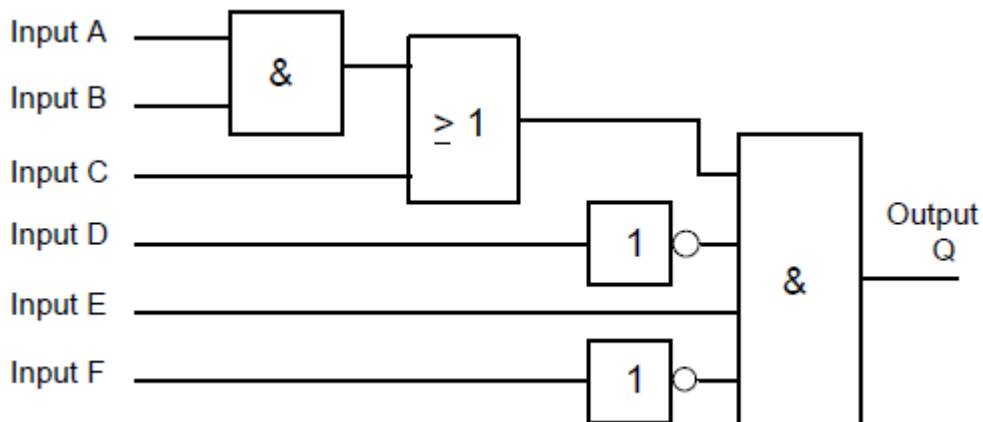
Figure 5.30 Ladder diagram and equivalent function block diagram

# Boolean algebra

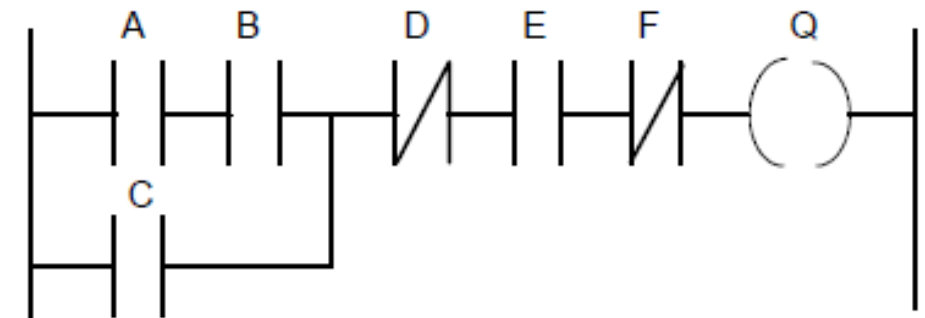
- AND operation  $A.B = Q$
- OR operation  $A + B = Q$
- NOT operation  $\bar{A} = Q$
- $A + B.C = Q$ 
  - Mitsubishi notation:  $X400 + X401.X402 = Y430$
  - Siemens notation:  $I0.0 + I0.1.I0.2 = Q2.0$
- $A + \bar{B} = Q$



$$A. \bar{B} + \bar{A}.B = Q$$

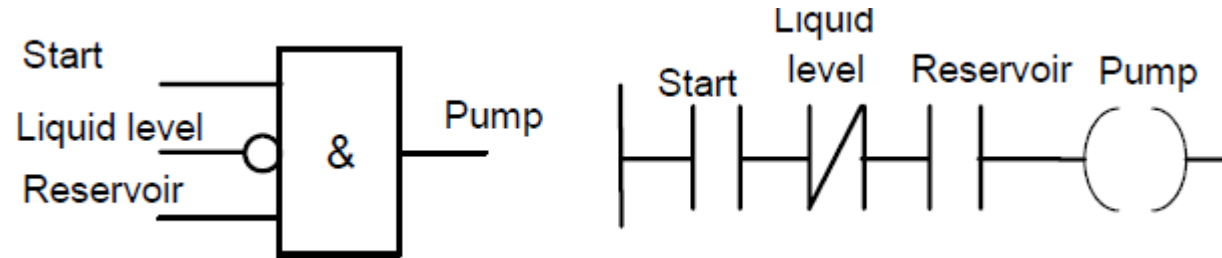


$$(A.B + C). \bar{D}.E. \bar{F} = Q$$

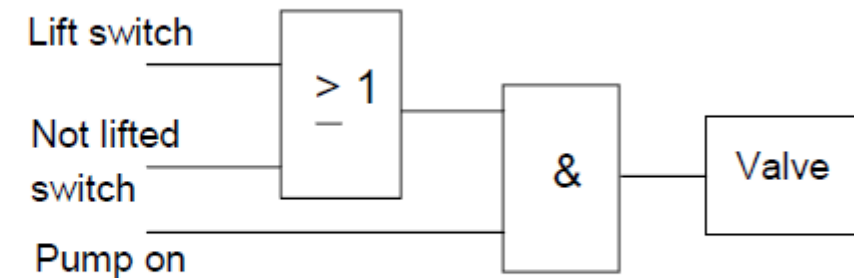
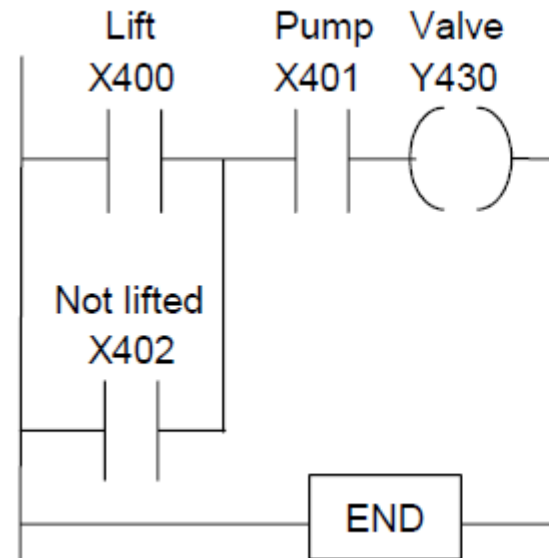


# PLC diagram Example:

- The level of liquid in the tank is below the required level and there is liquid in the reservoir from which it is to be pumped

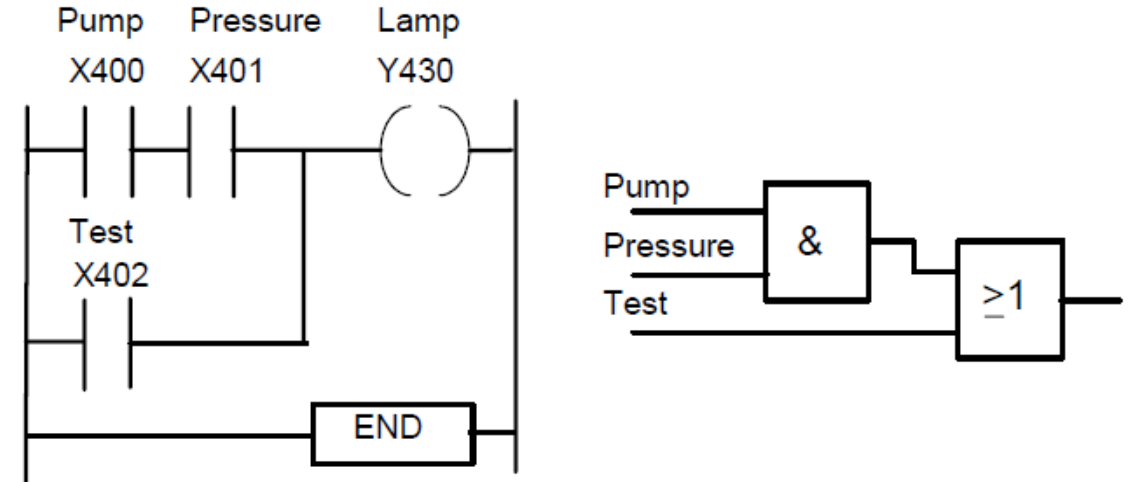


- A valve which is to be operated to lift a load when a pump is running and either the lift switch is operated or a switch operated indicating that the load has not already been lifted and is at the bottom of its lift channel

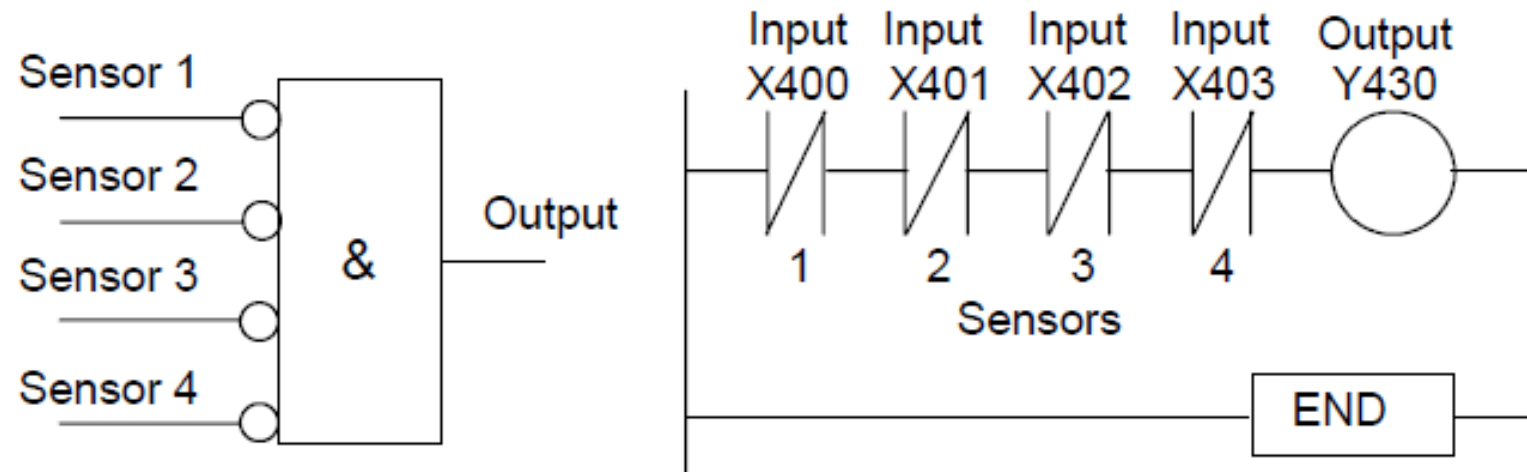


# PLC diagram Example:

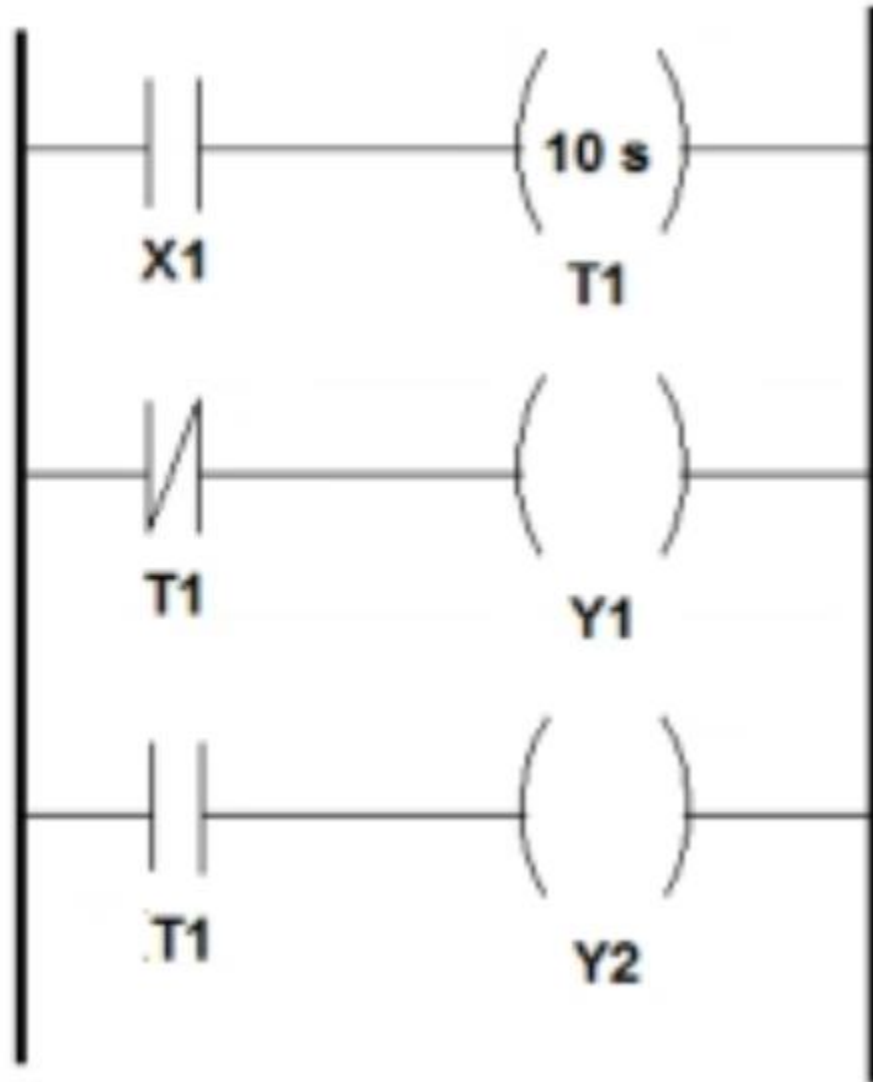
- A signal lamp is required to be switched on if a pump is running and the pressure is satisfactory, or if the lamp test switch is closed



- A system where there has to be no output when any one of four sensors gives an output

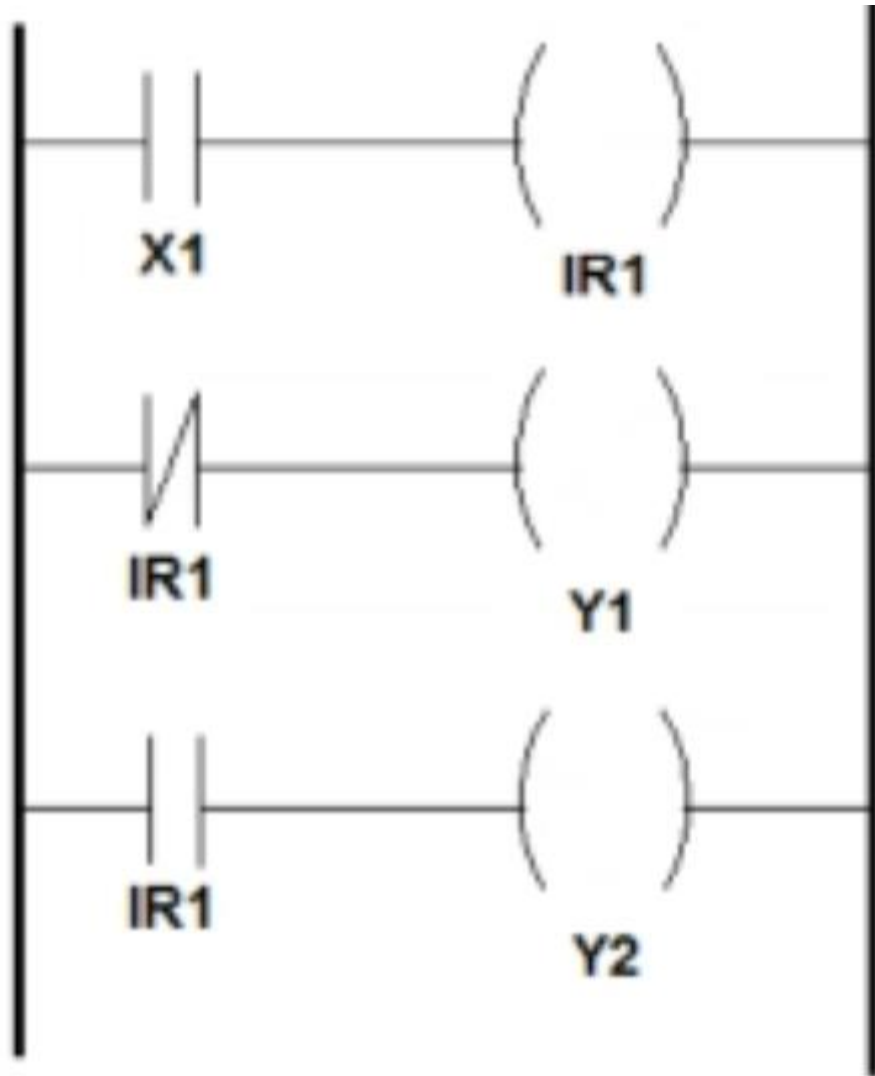


# Timer



- Timer having 2 components
  - First component gets triggered when signal comes
  - Second component may be NO or NC switch
- Both components having same identifier
- Once connection broken to the Timer, gets reset

# Internal Relay or Marker

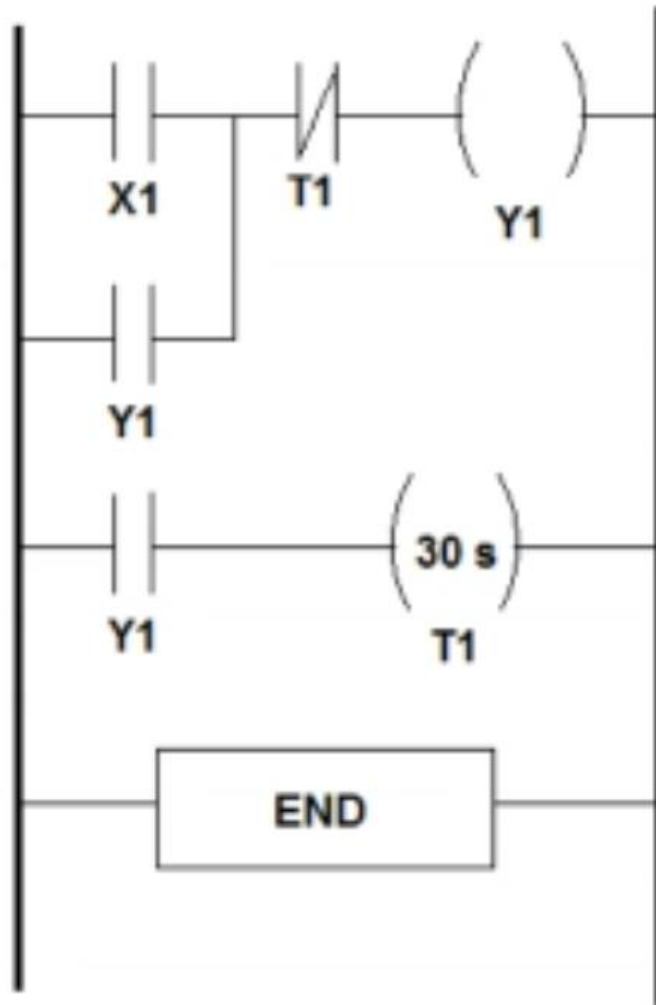


- Internal Relay having 2 components
  - First component gets triggered when signal comes
  - Second component may be set of NO or NC switch
- Both components having same identifier
- Once connection broken to the Internal Relay, gets de-energized



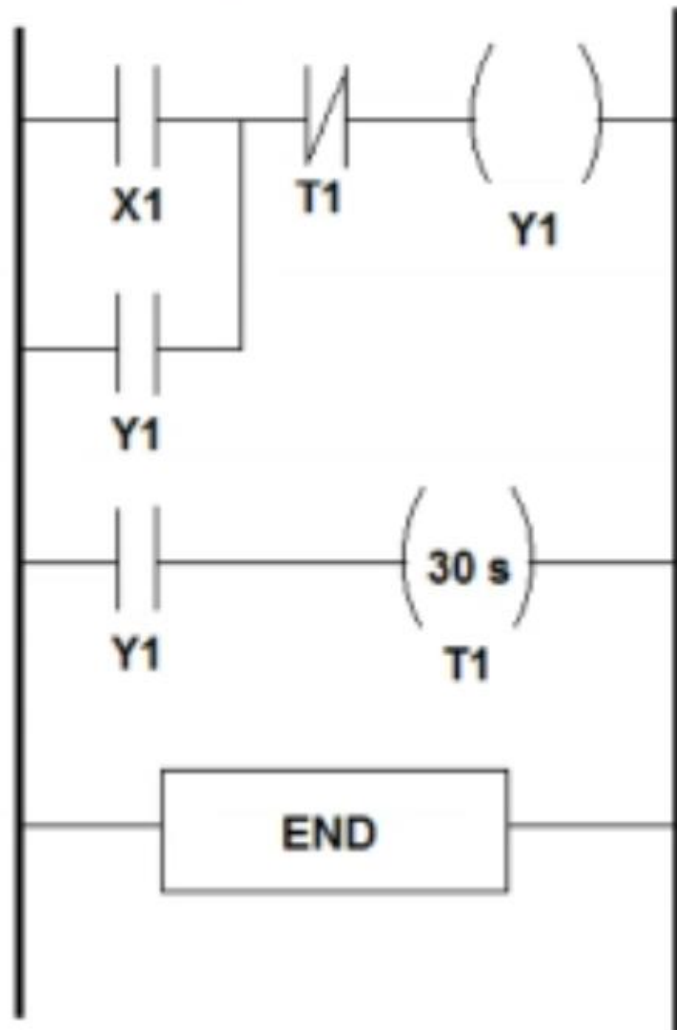
# PLC programming example:

When a push-button switch is pressed, a motor starts running. It runs for 30 seconds and then stops.



# PLC programming example:

When a push-button switch is pressed, a motor starts running. It runs for 30 seconds and then stops.



X1 - start button

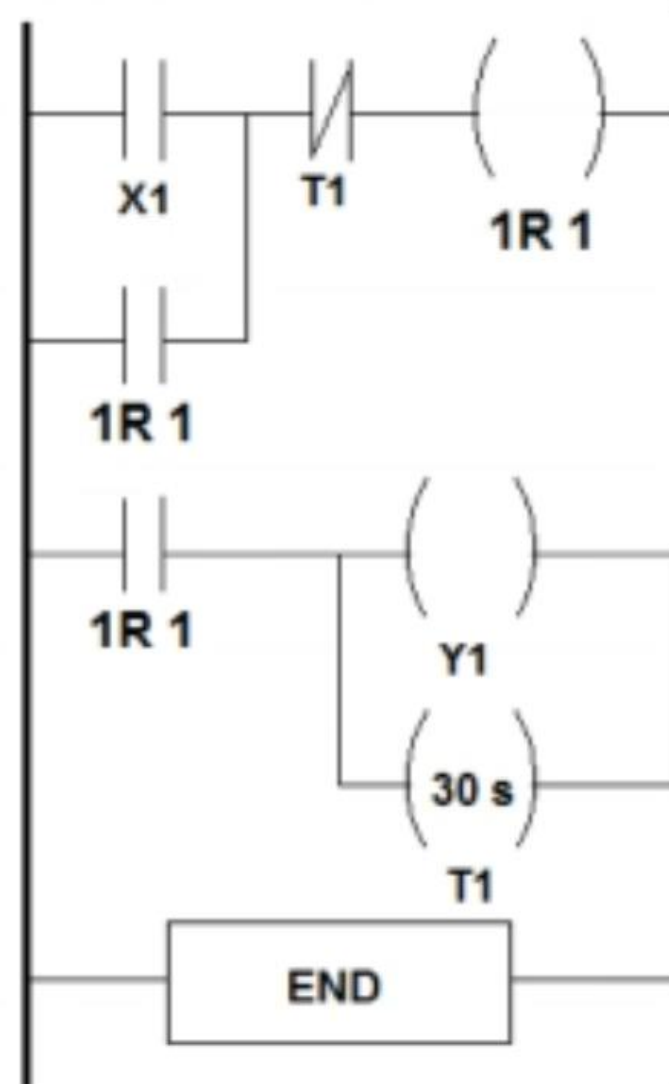
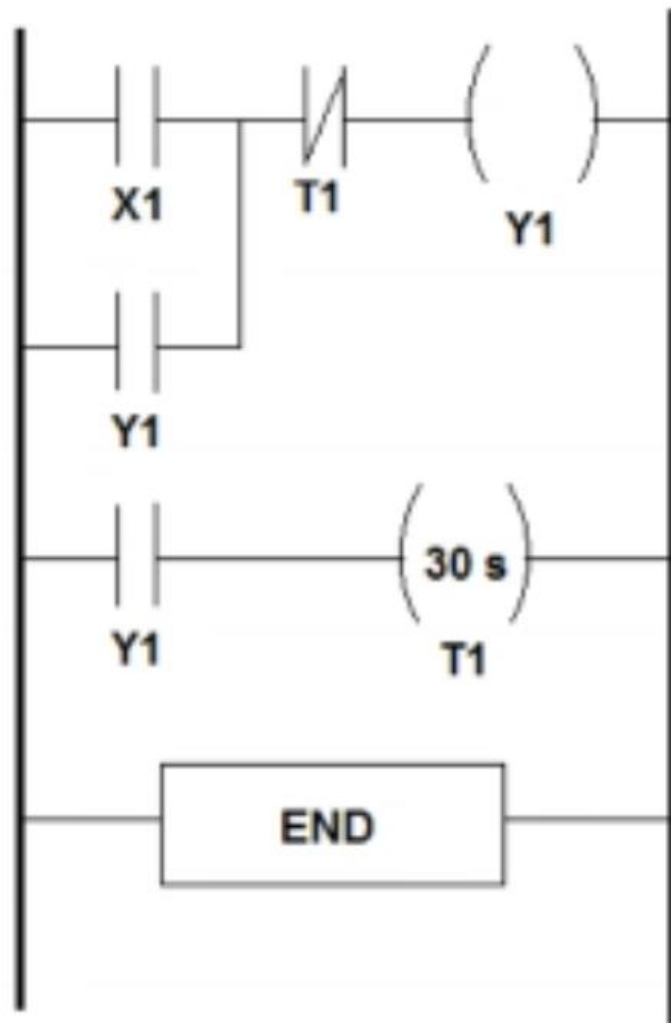
Y1 - motor

T1 is the timer. It is internal.



# PLC programming example:

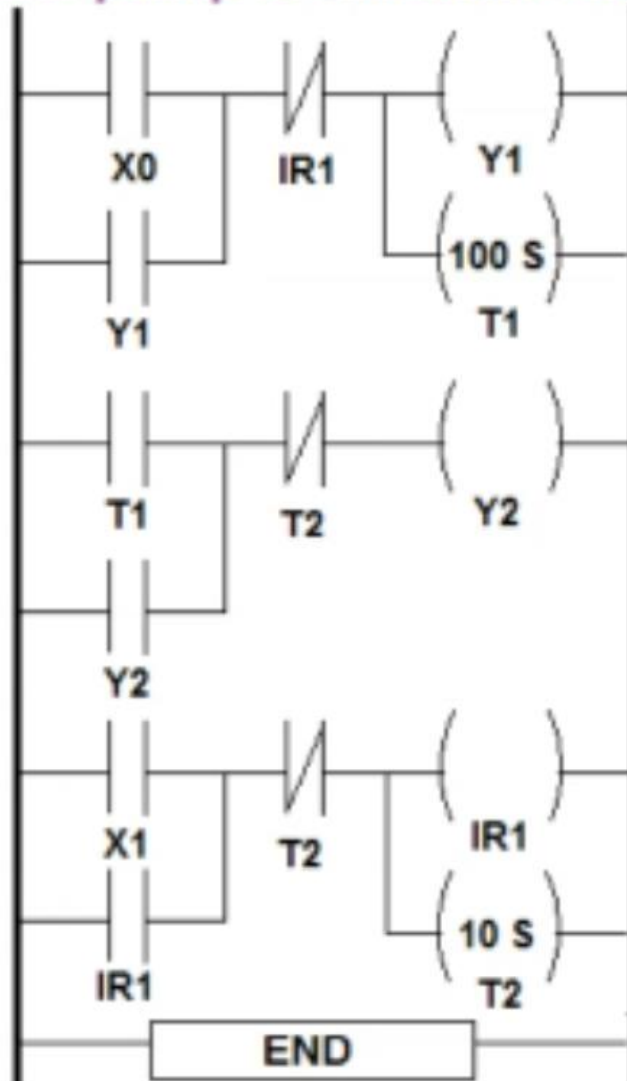
When a push-button switch is pressed, a motor starts running. It runs for 30 seconds and then stops.



I

# PLC programming example:

Devise a circuit that can be used to start a motor and then after a delay of 100 s start a pump. When the motor is switched off there should be a delay of 10 s before the pump is switched off.



X0 - START BUTTON

X1 - STOP BUTTON

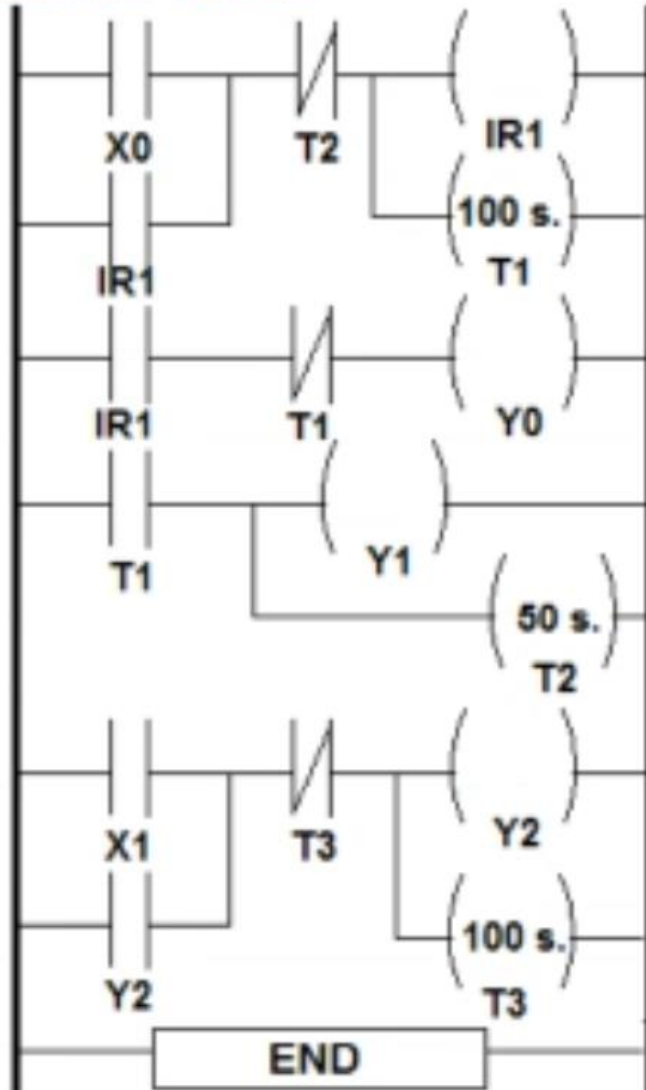
Y1 - MOTOR

Y2 - PUMP

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The timers T1 and T2 and the internal relay IR1 and internal to the PLC.

# PLC programming example:

Device a circuit that could be used with a domestic washing machine to switch on a pump to pump water for 100 s into the machine, then switch off and switch on a heater for 50 s to heat the water. The heater is then switched off. Another pump is to empty the water from the machine for 100 s.



X0 - START PUMP TO FILL  
X1 - START PUMP TO EMPTY  
Y0 - PUMP FOR FILLING  
Y1 - HEATER  
Y2 - PUMP TO EMPTY

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TIMERS T1, T2, and T3  
INTERNAL RELAY IR1

INTERNAL